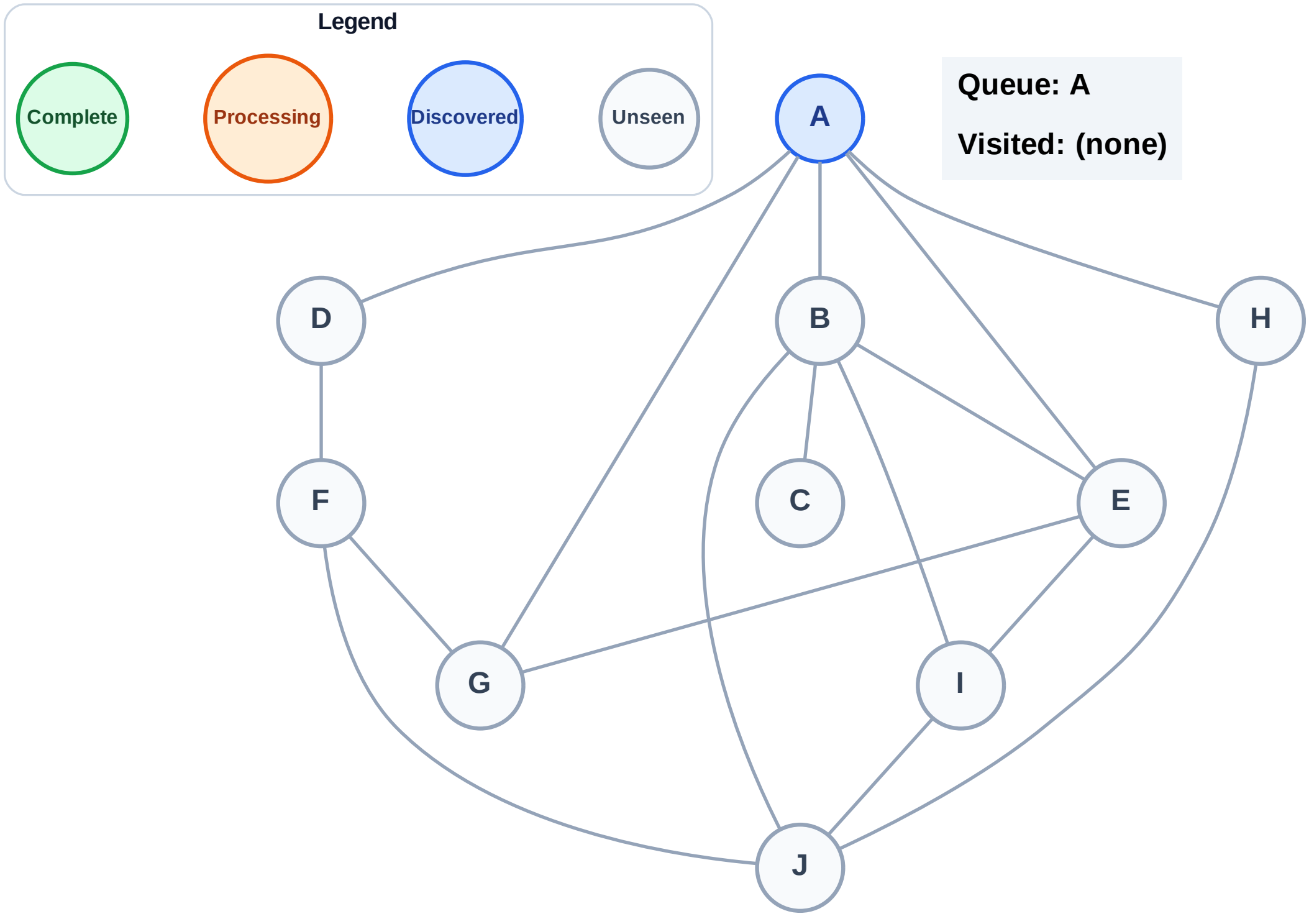
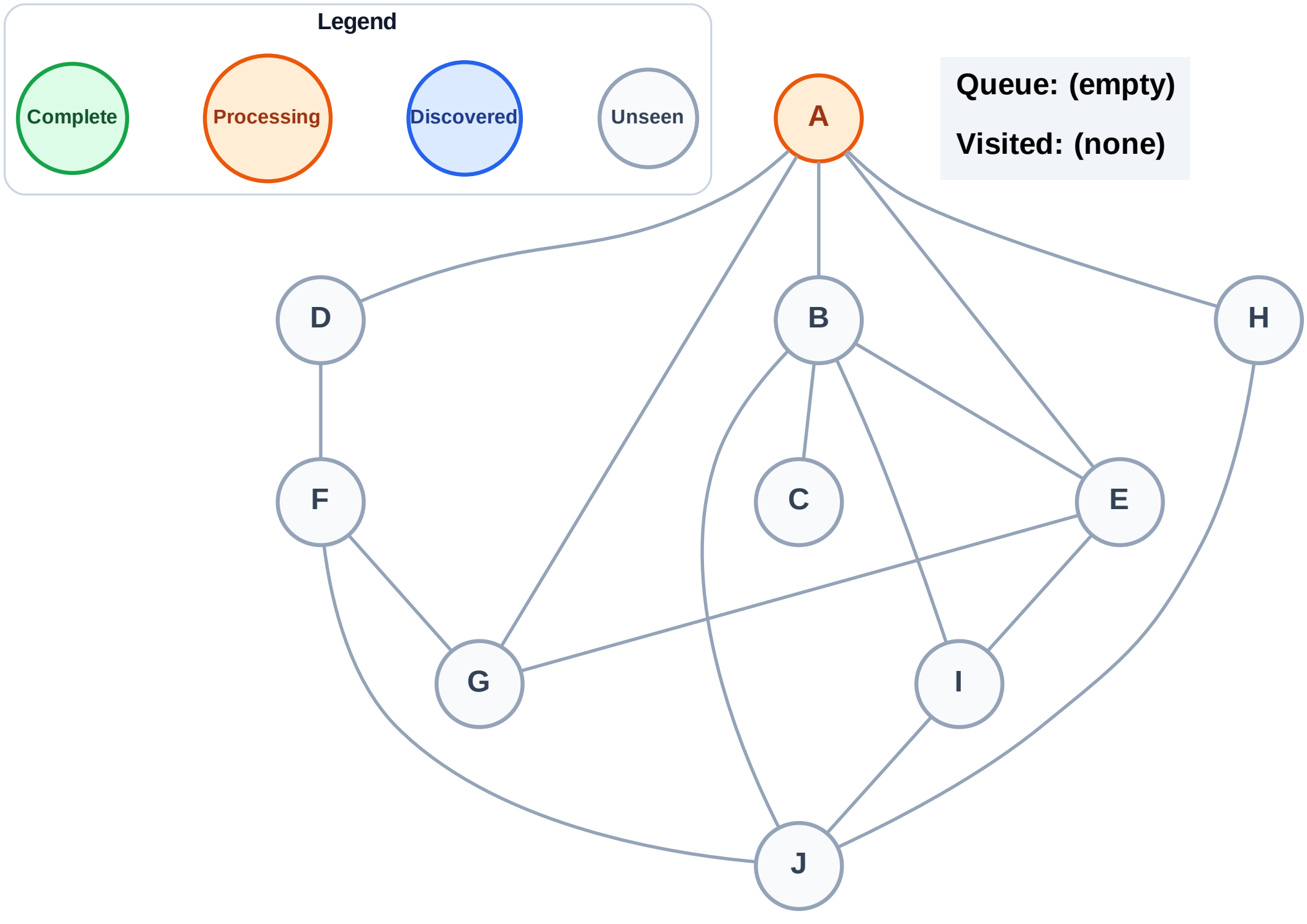
BFS Traversal

Step 1: Start at A



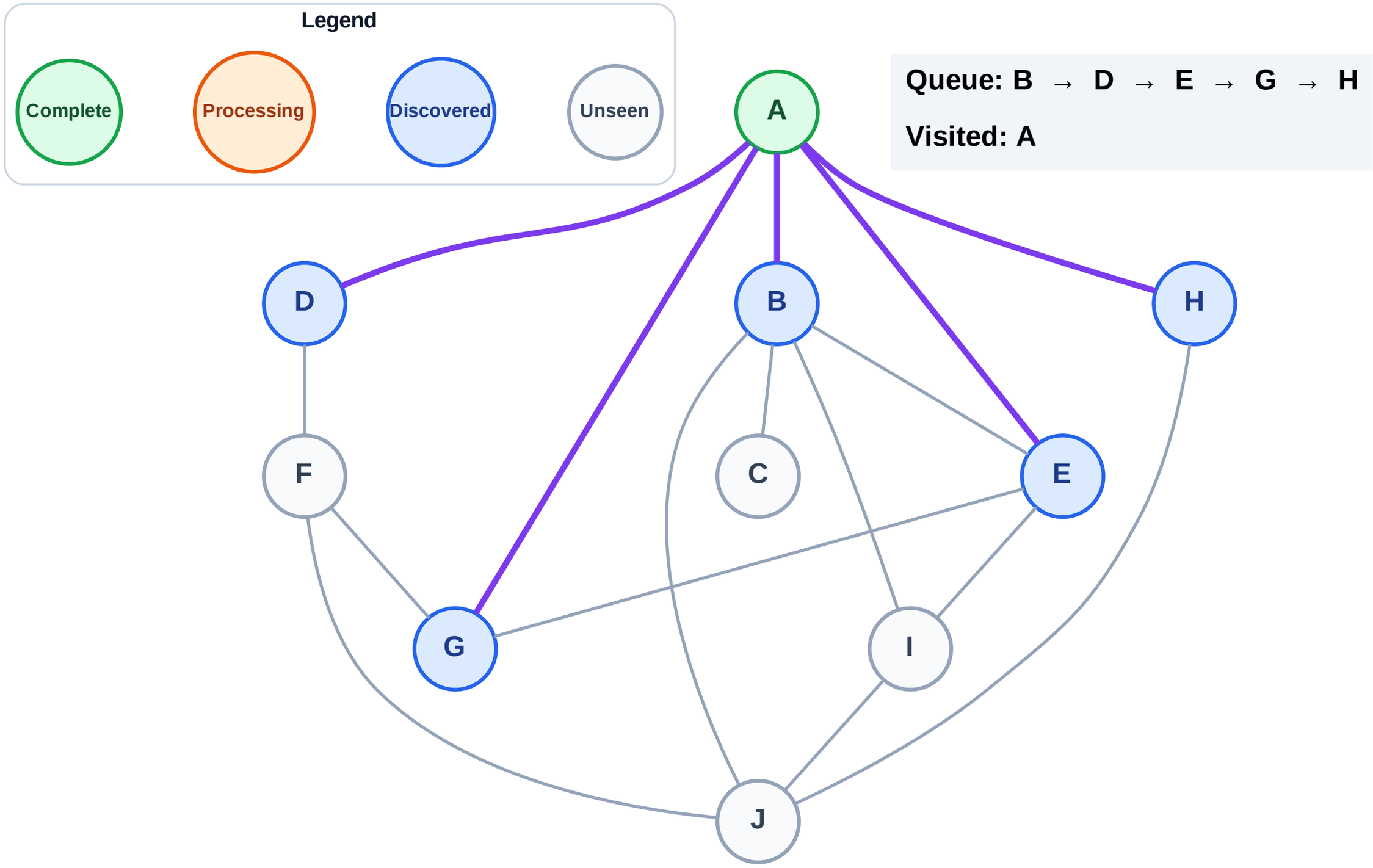
BFS Traversal

Step 2: Process node A



BFS Traversal

Step 3: Discover B, D, E, G, H from A

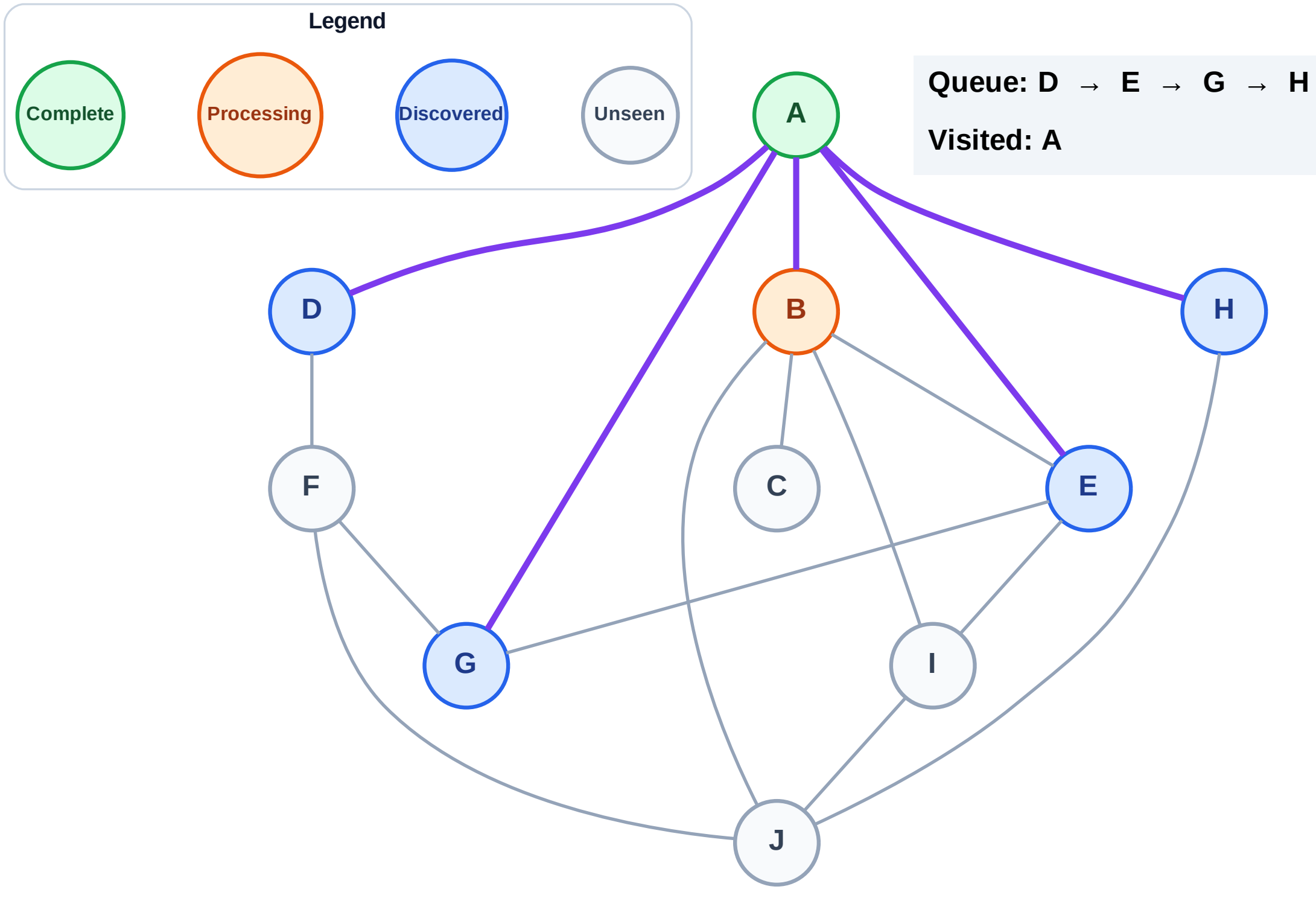


Queue: B → D → E → G → H

Visited: A

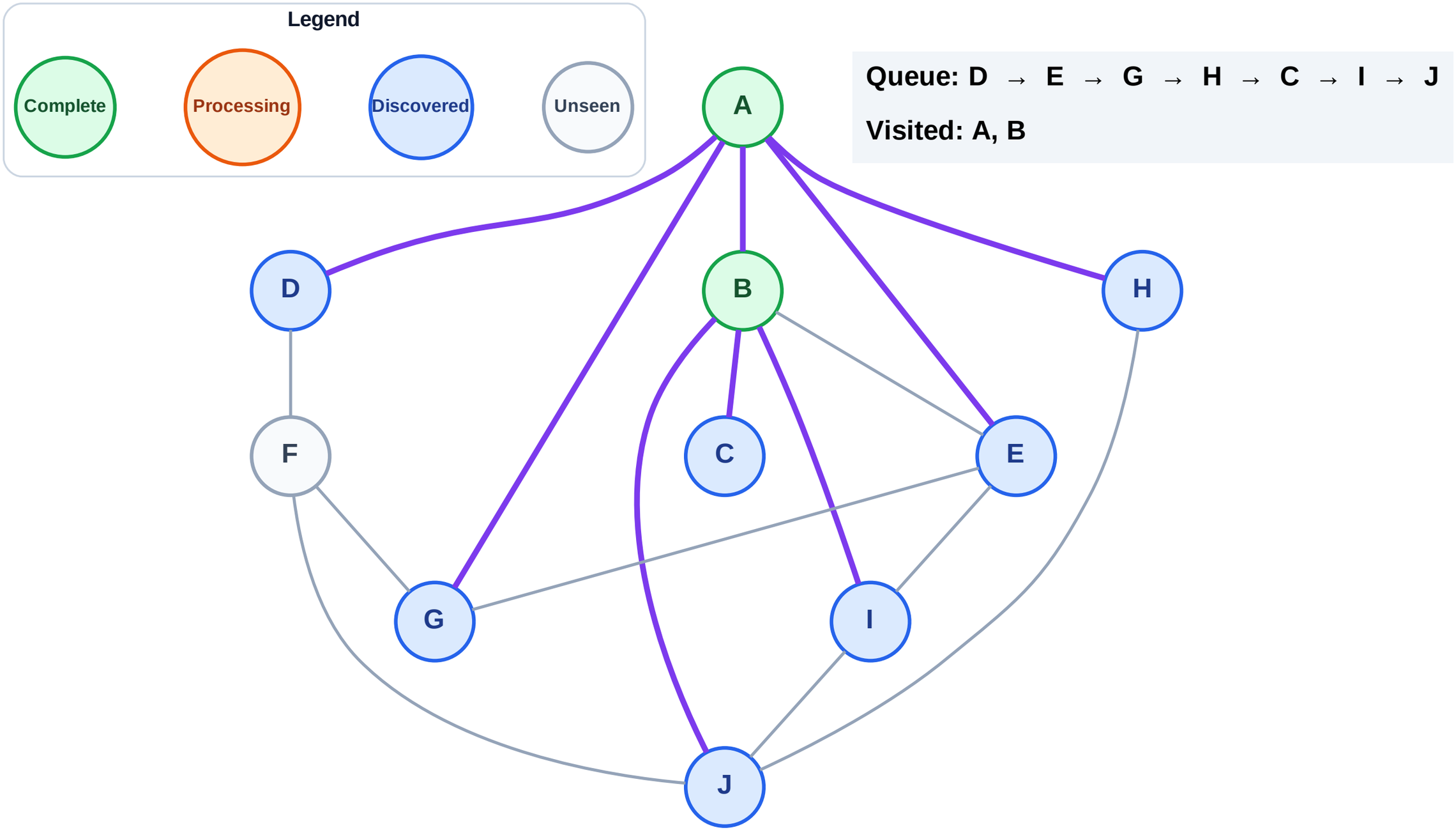
BFS Traversal

Step 4: Process node B



BFS Traversal

Step 5: Discover C, I, J from B

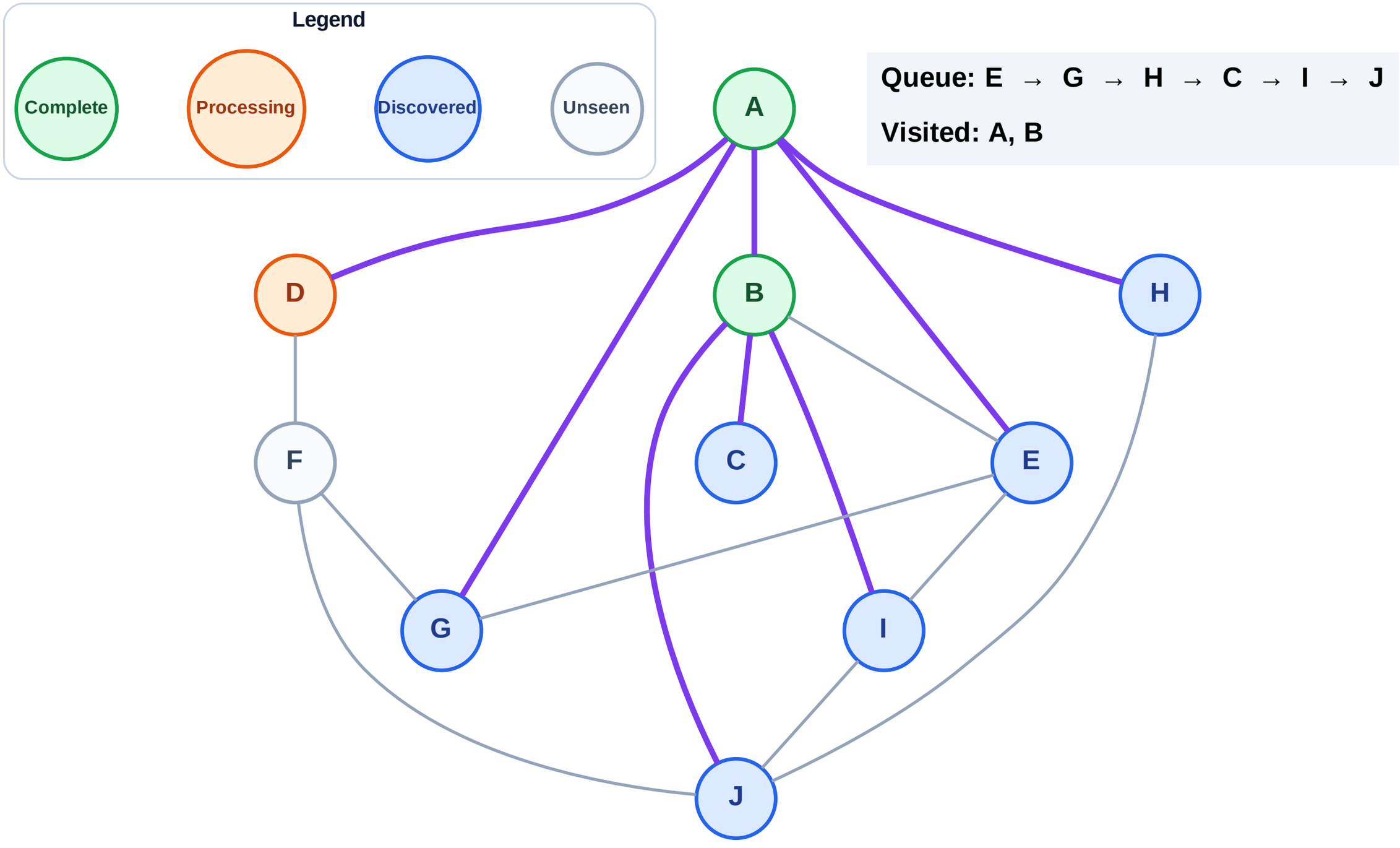


Queue: D → E → G → H → C → I → J

Visited: A, B

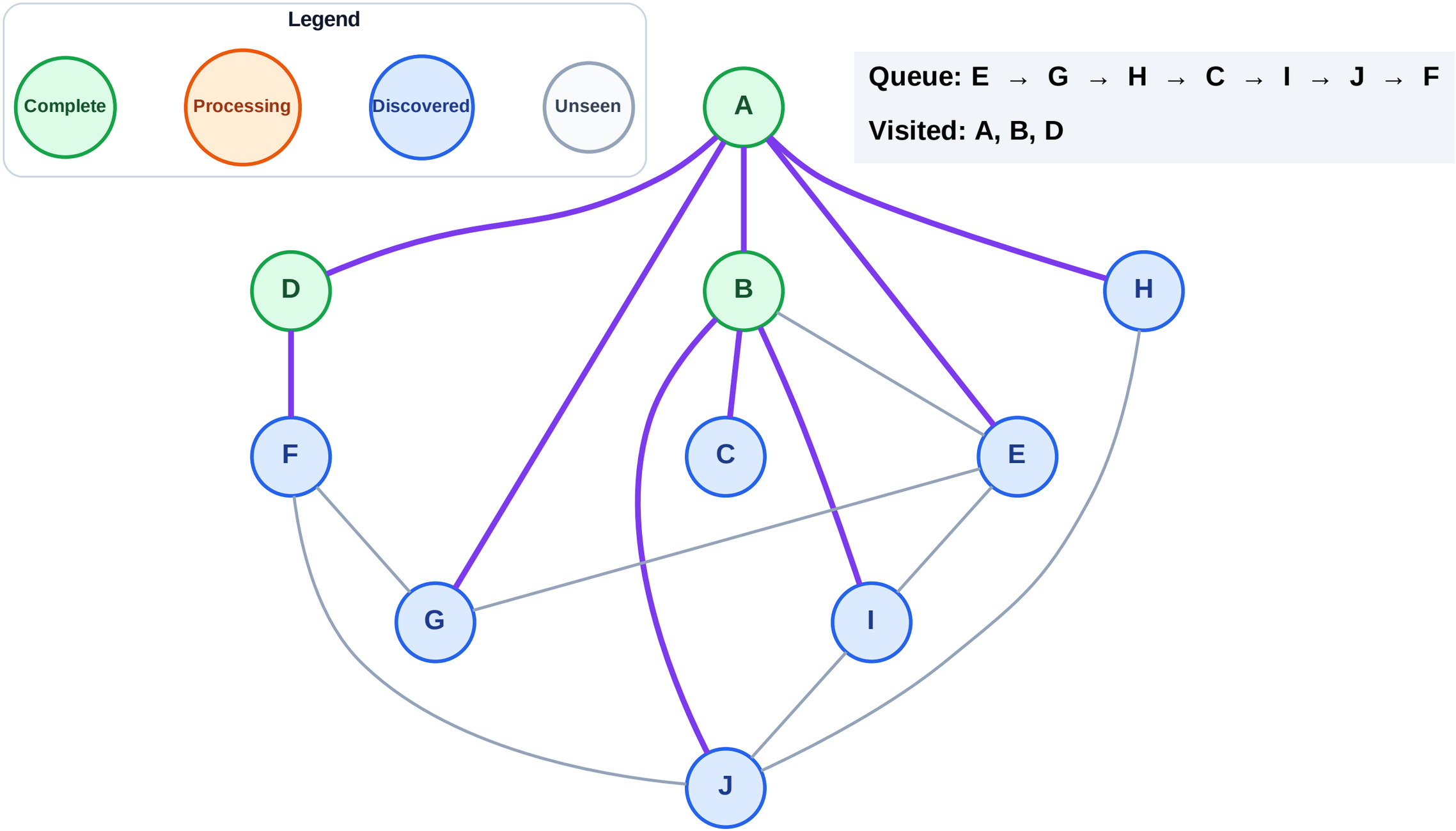
BFS Traversal

Step 6: Process node D



BFS Traversal

Step 7: Discover F from D

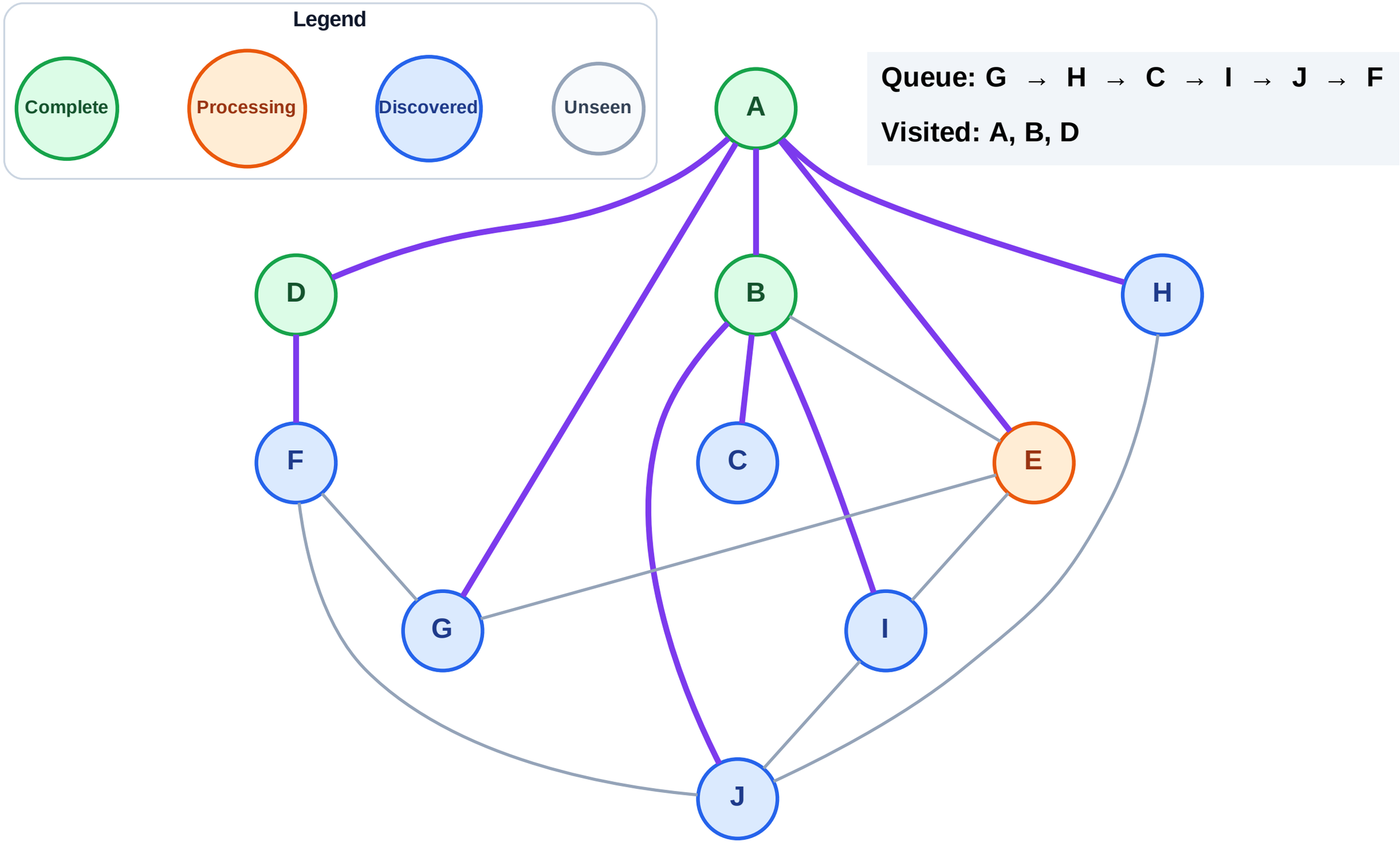


Queue: E → G → H → C → I → J → F

Visited: A, B, D

BFS Traversal

Step 8: Process node E

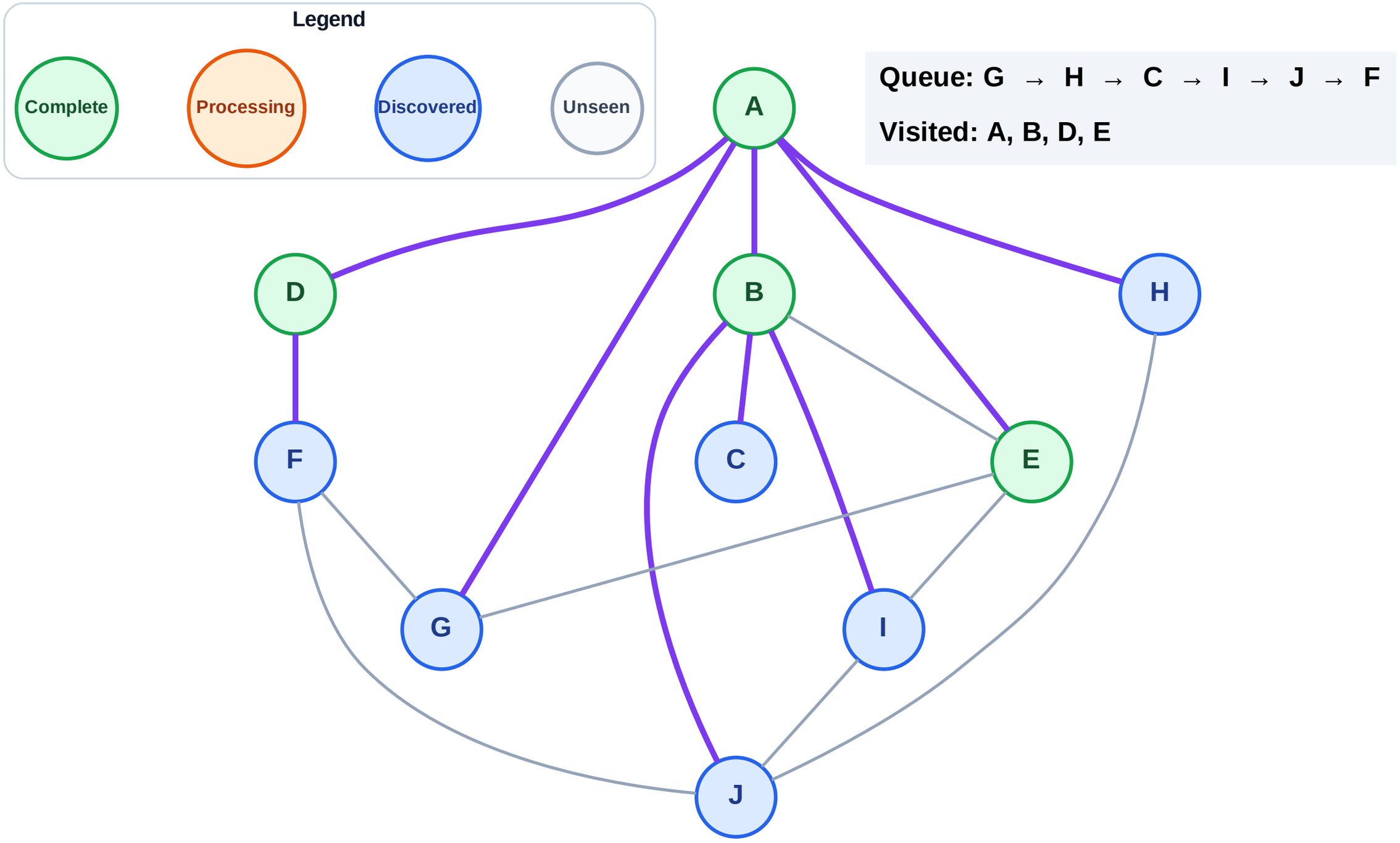


Queue: G → H → C → I → J → F

Visited: A, B, D

BFS Traversal

Step 9: No new neighbors from E

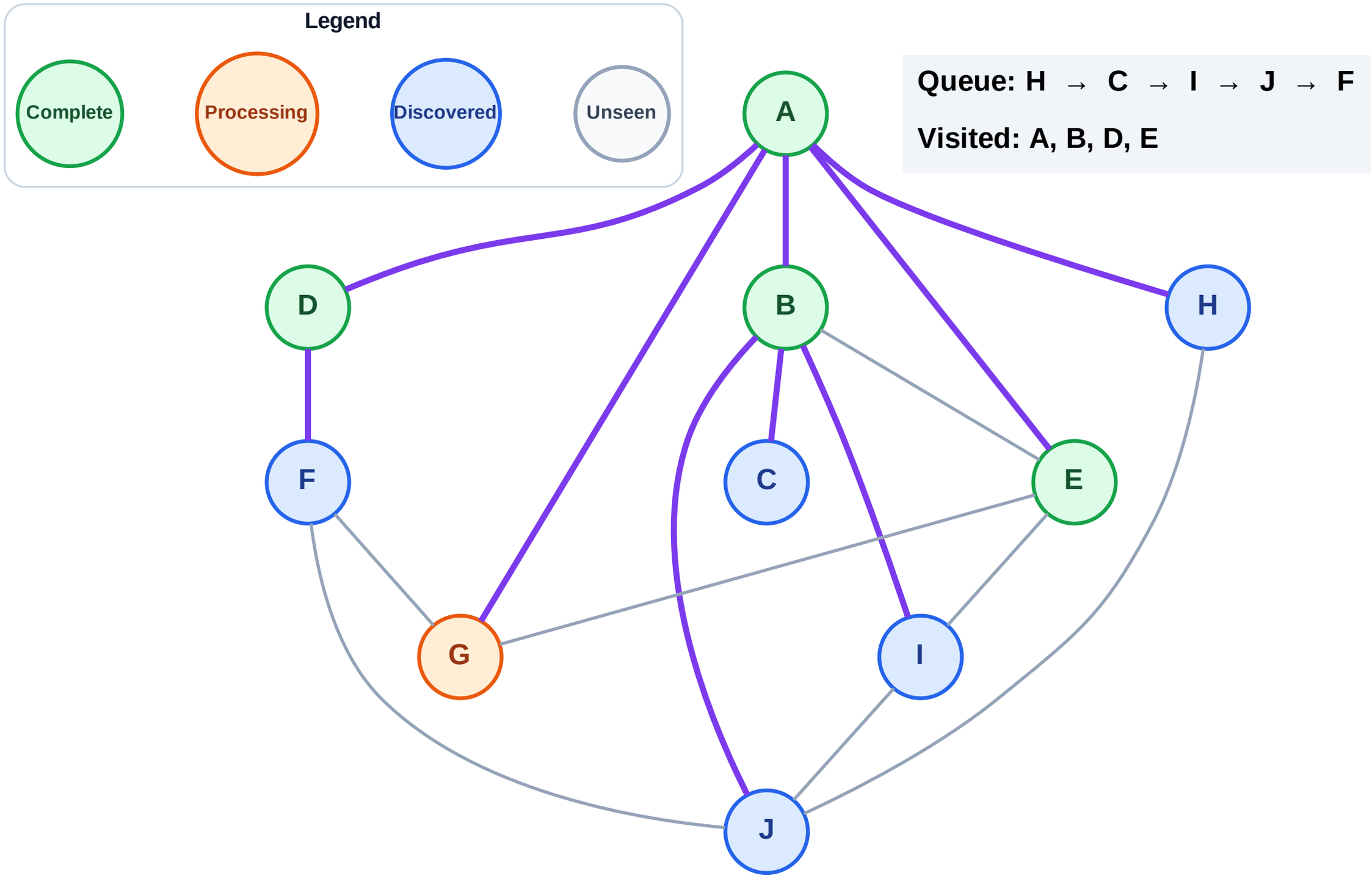


Queue: G → H → C → I → J → F

Visited: A, B, D, E

BFS Traversal

Step 10: Process node G

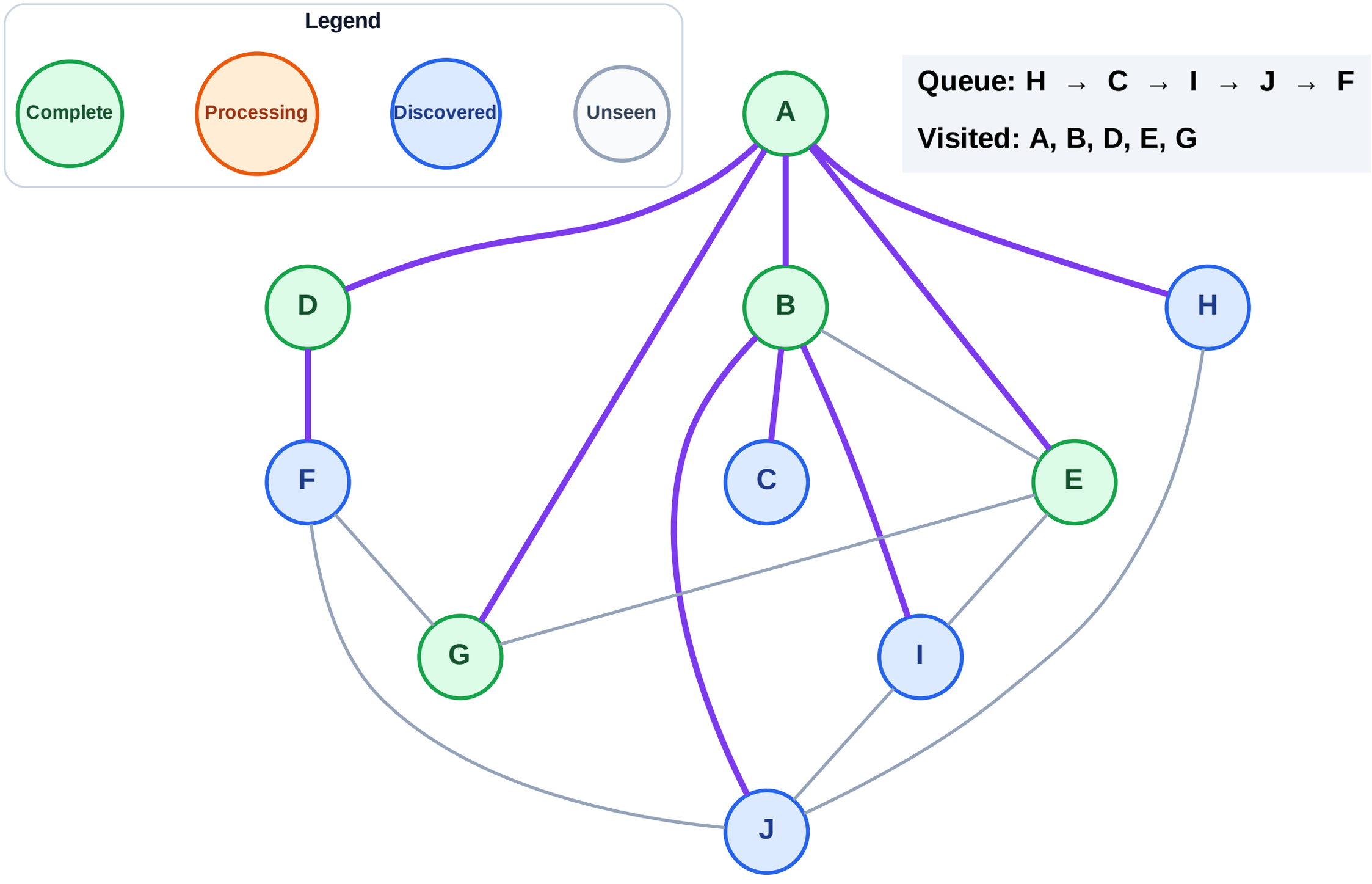


Queue: H → C → I → J → F

Visited: A, B, D, E

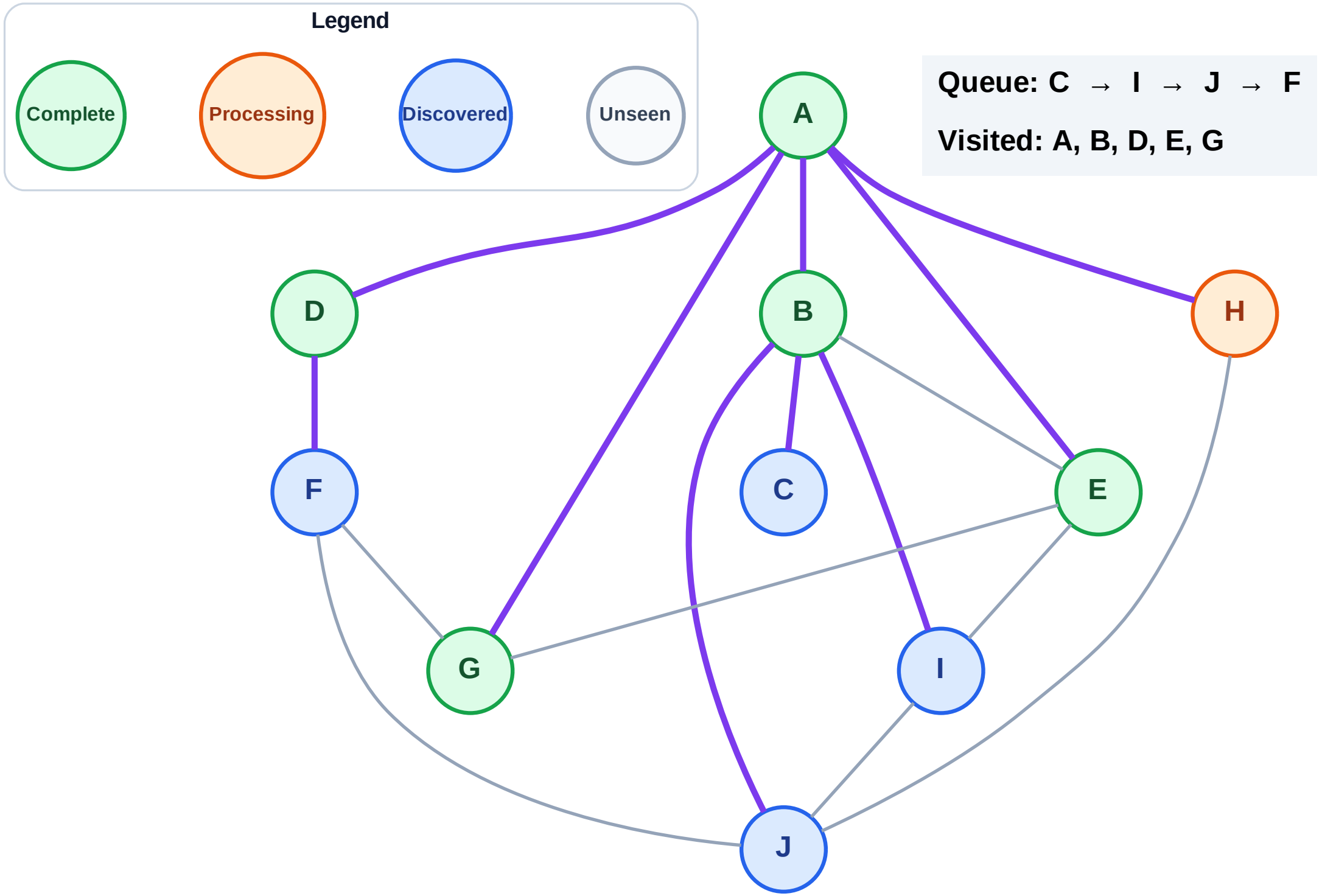
BFS Traversal

Step 11: No new neighbors from G



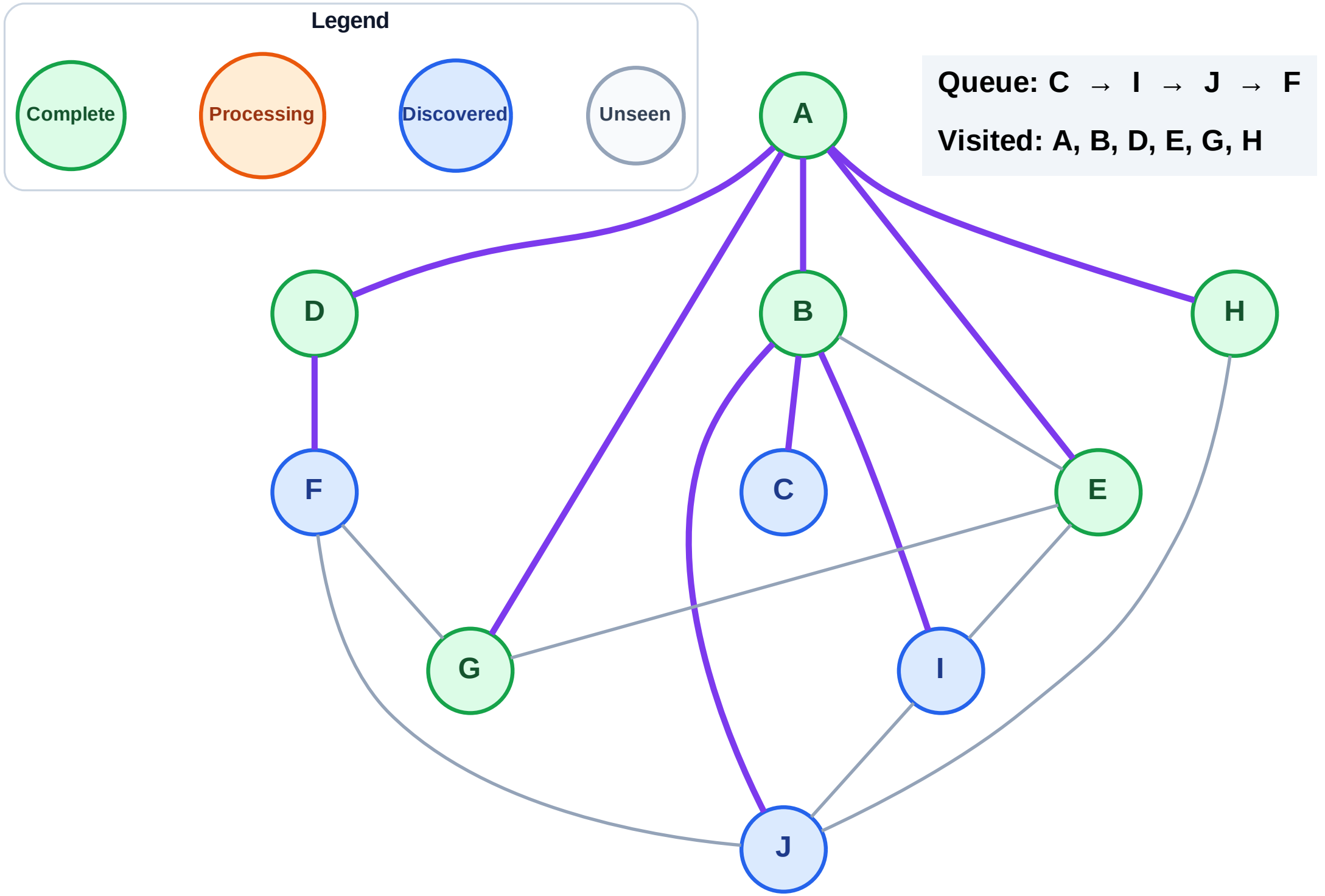
BFS Traversal

Step 12: Process node H



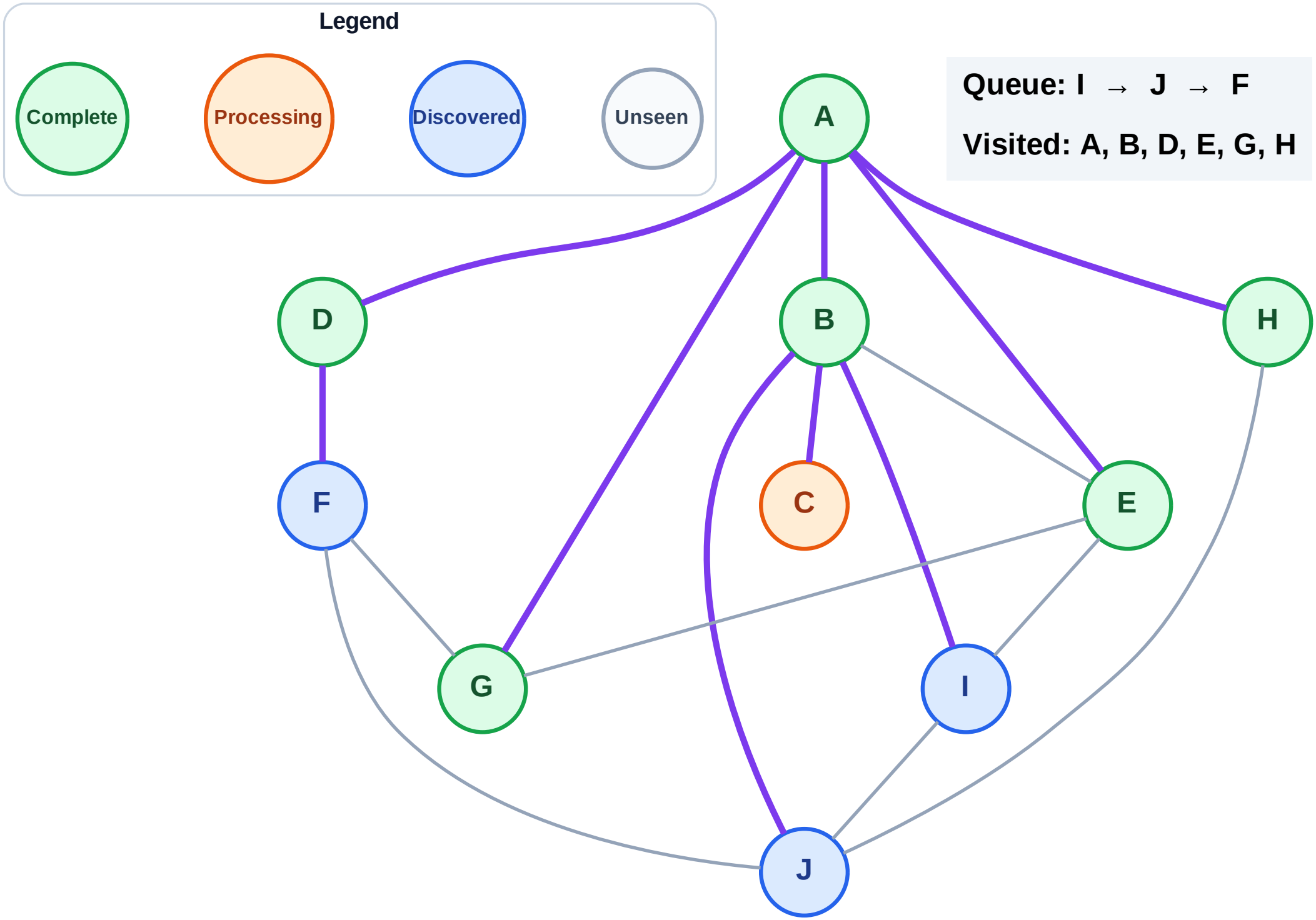
BFS Traversal

Step 13: No new neighbors from H



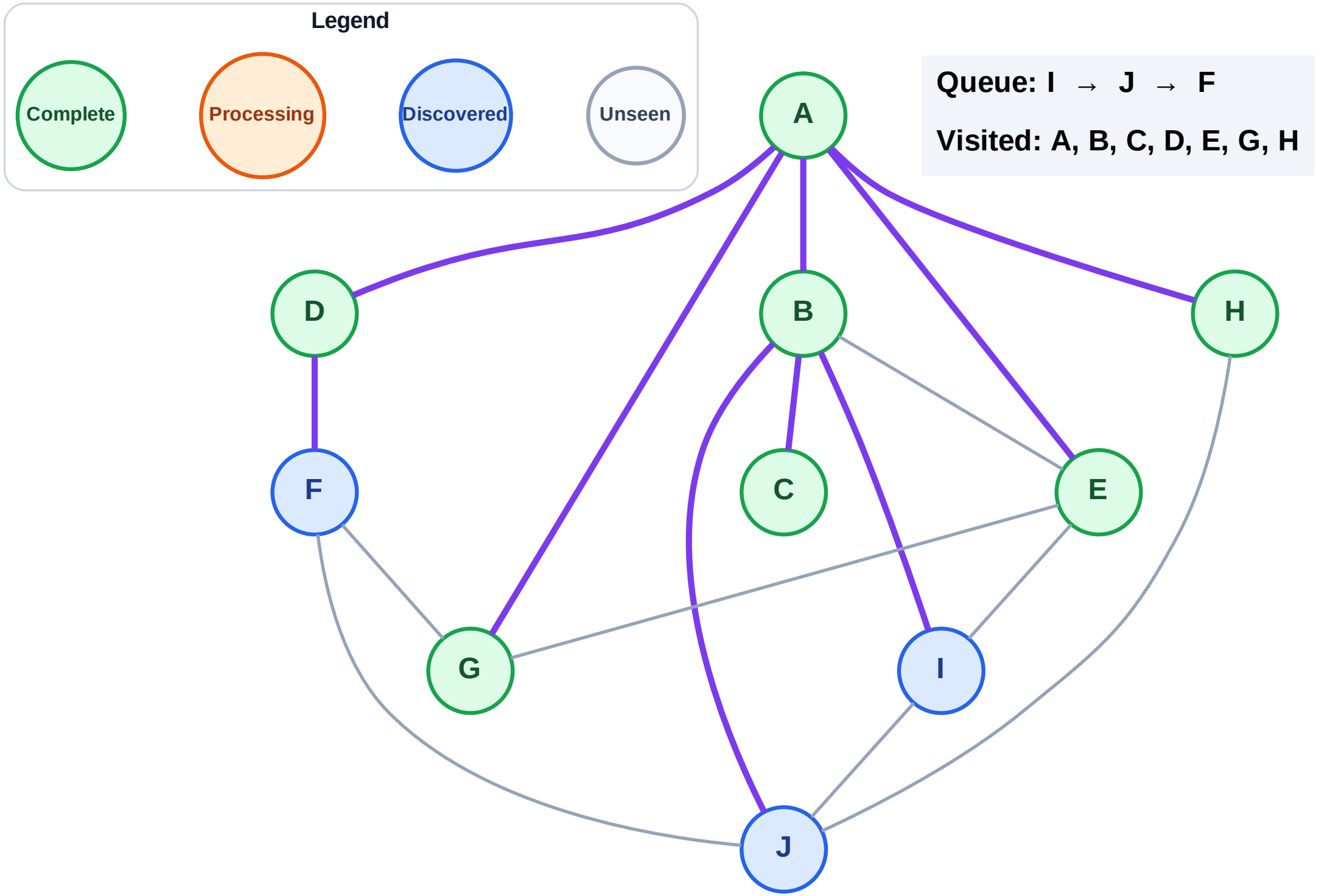
BFS Traversal

Step 14: Process node C



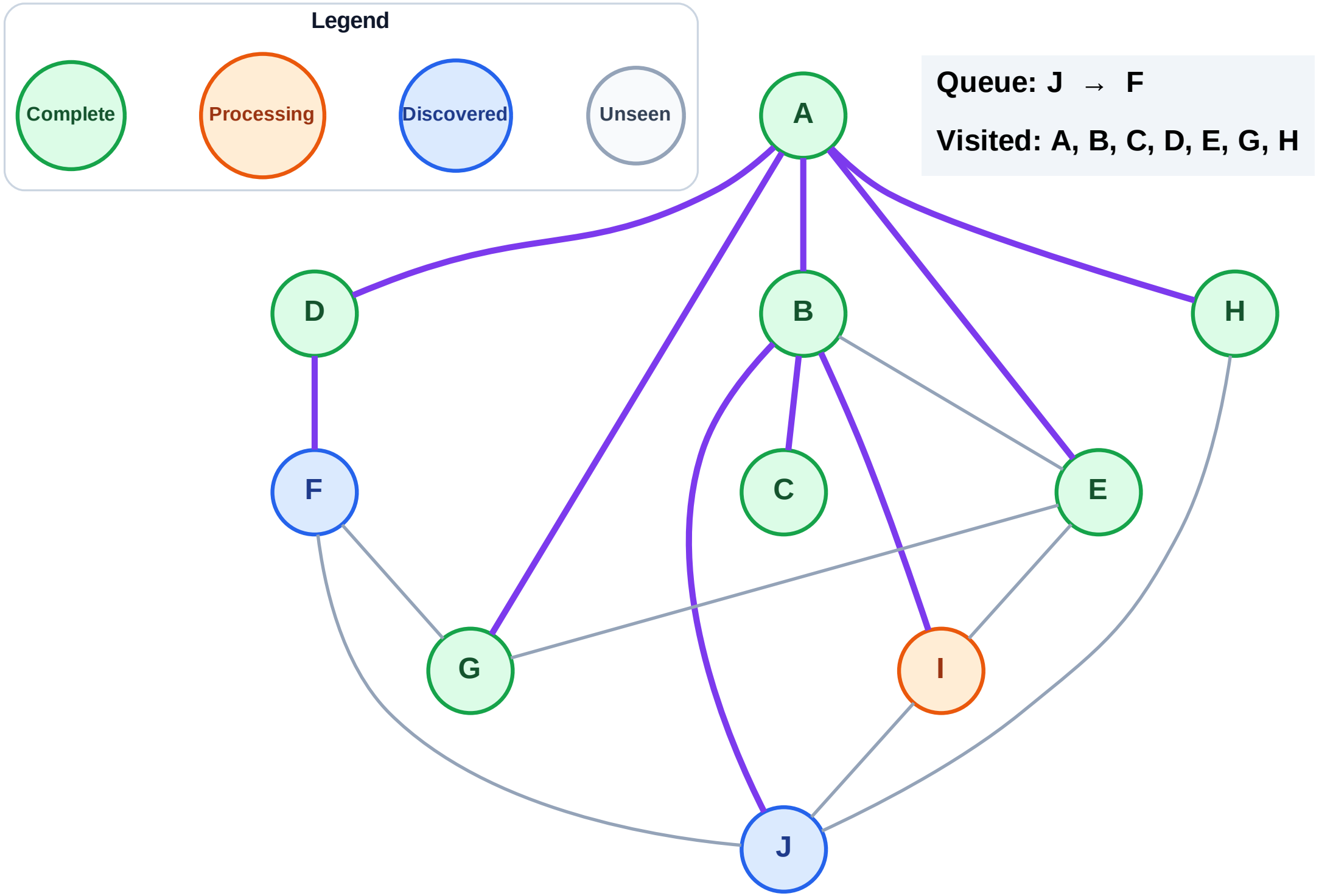
BFS Traversal

Step 15: No new neighbors from C



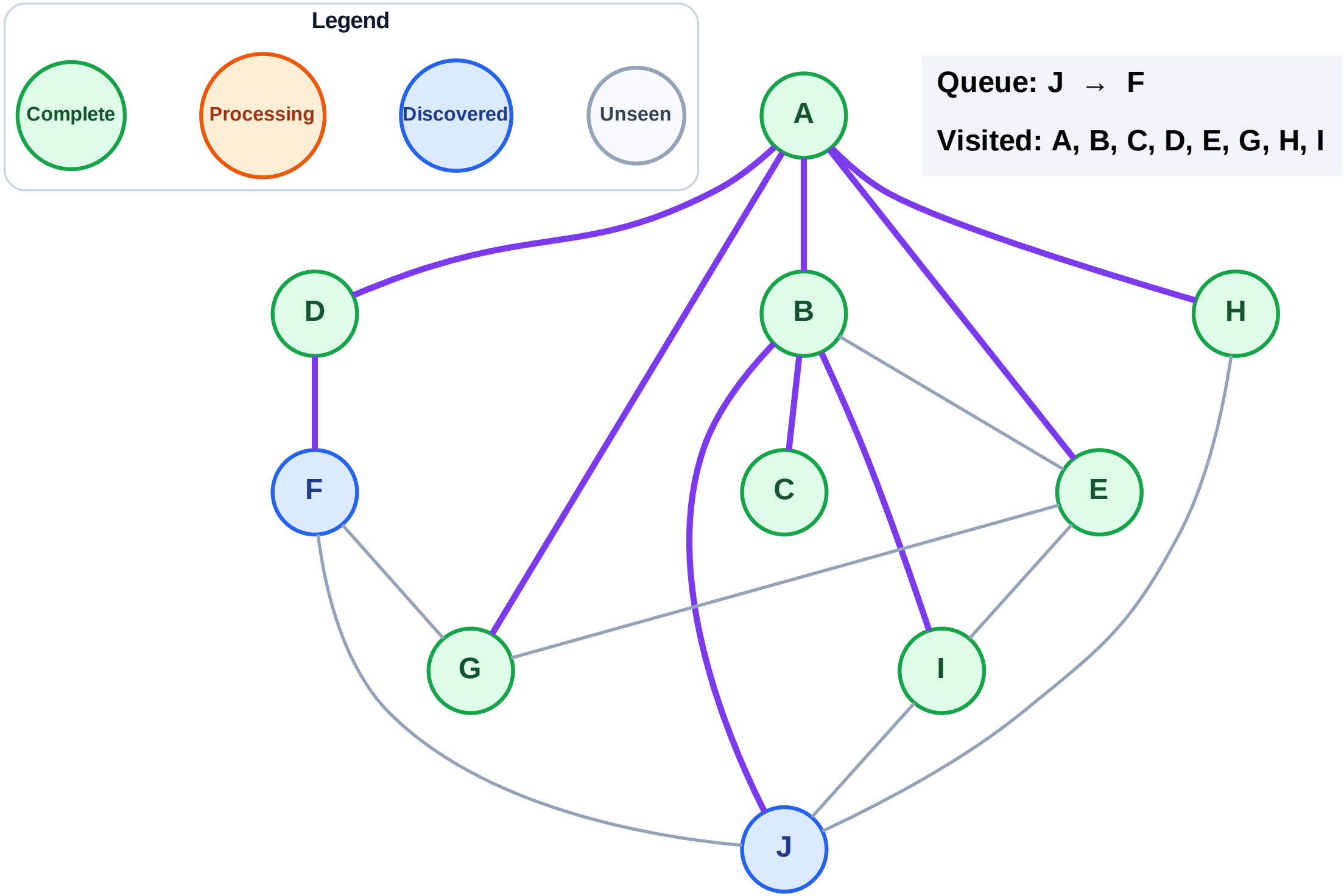
BFS Traversal

Step 16: Process node I



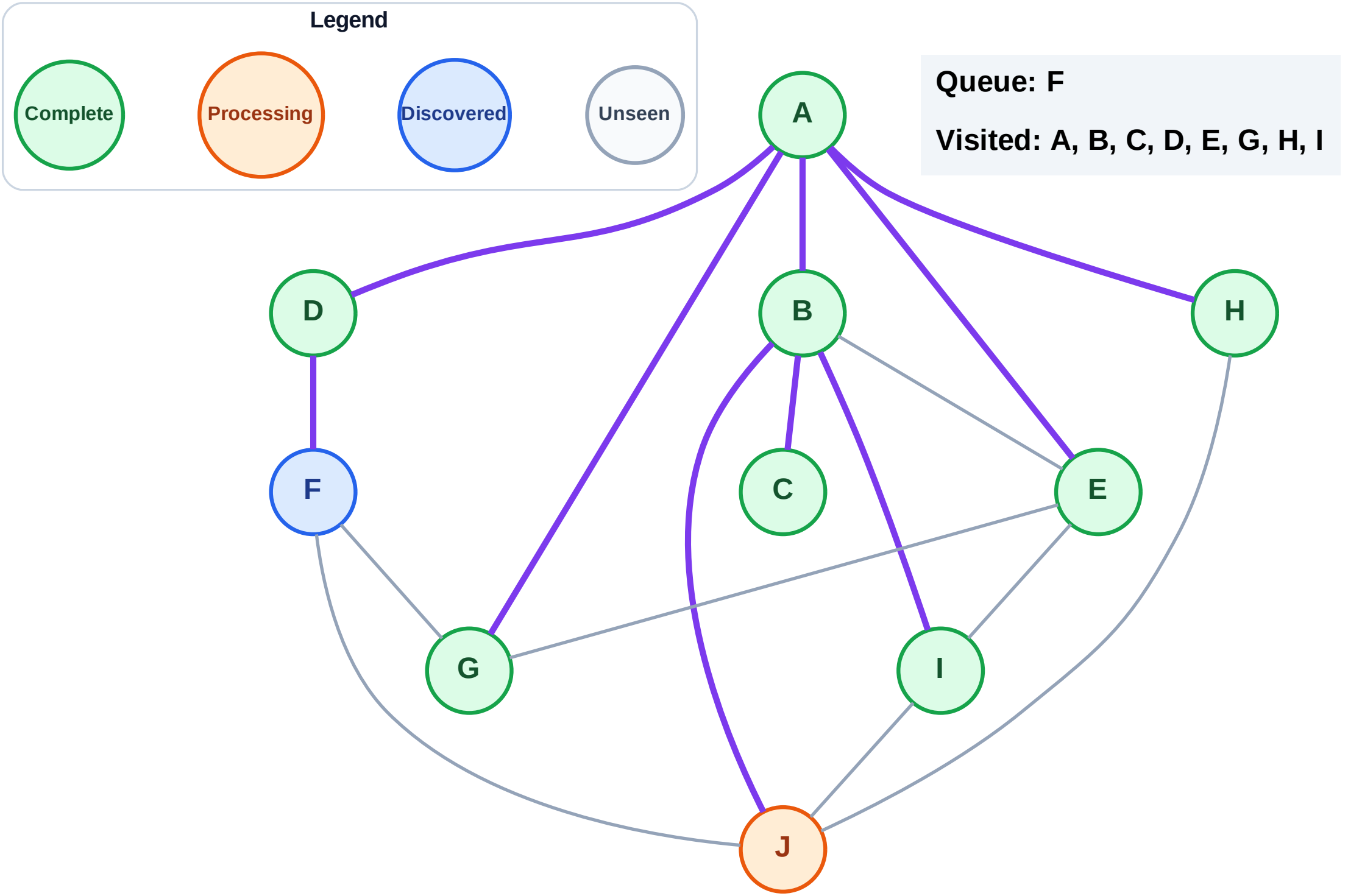
BFS Traversal

Step 17: No new neighbors from I



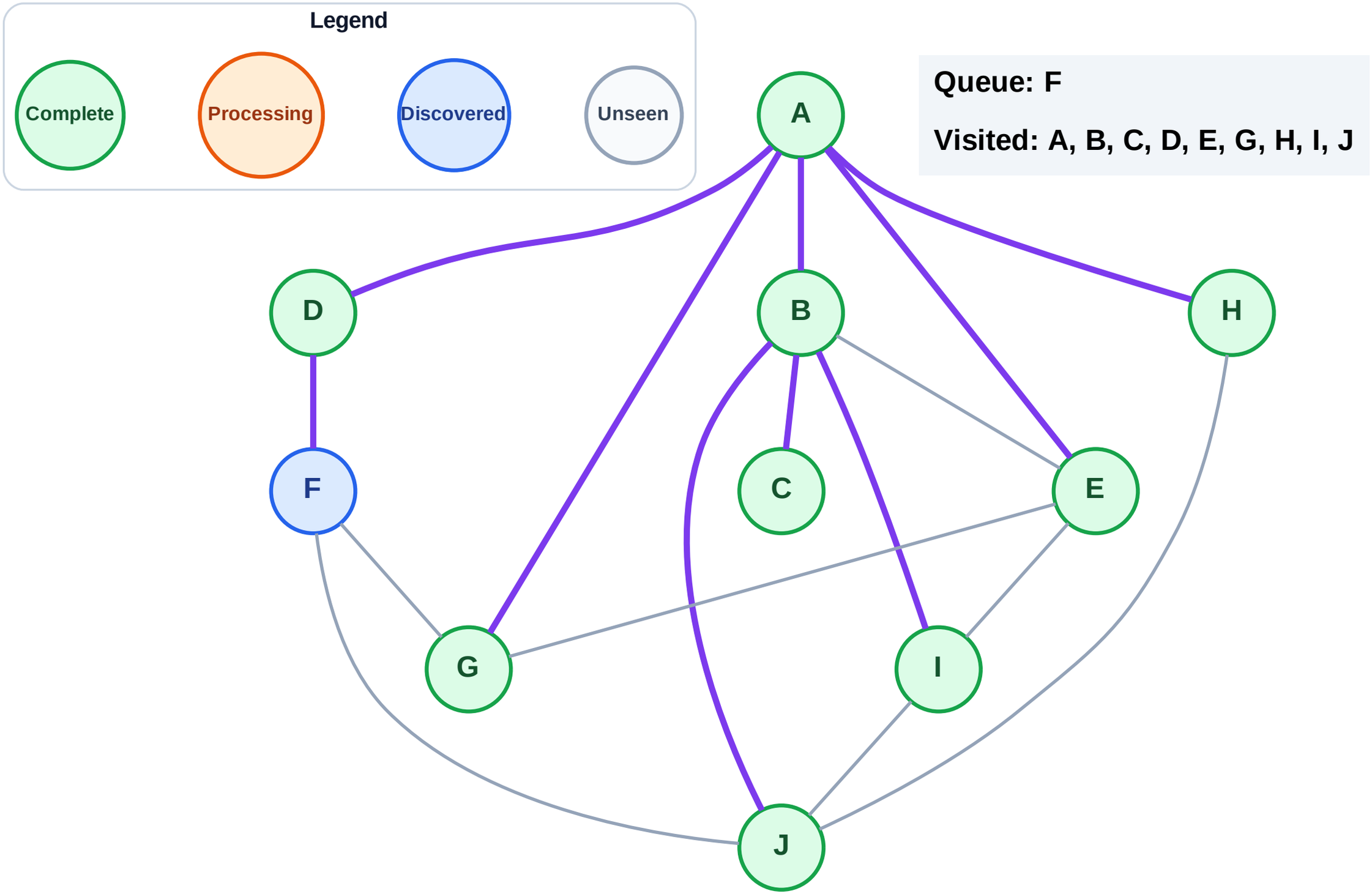
BFS Traversal

Step 18: Process node J



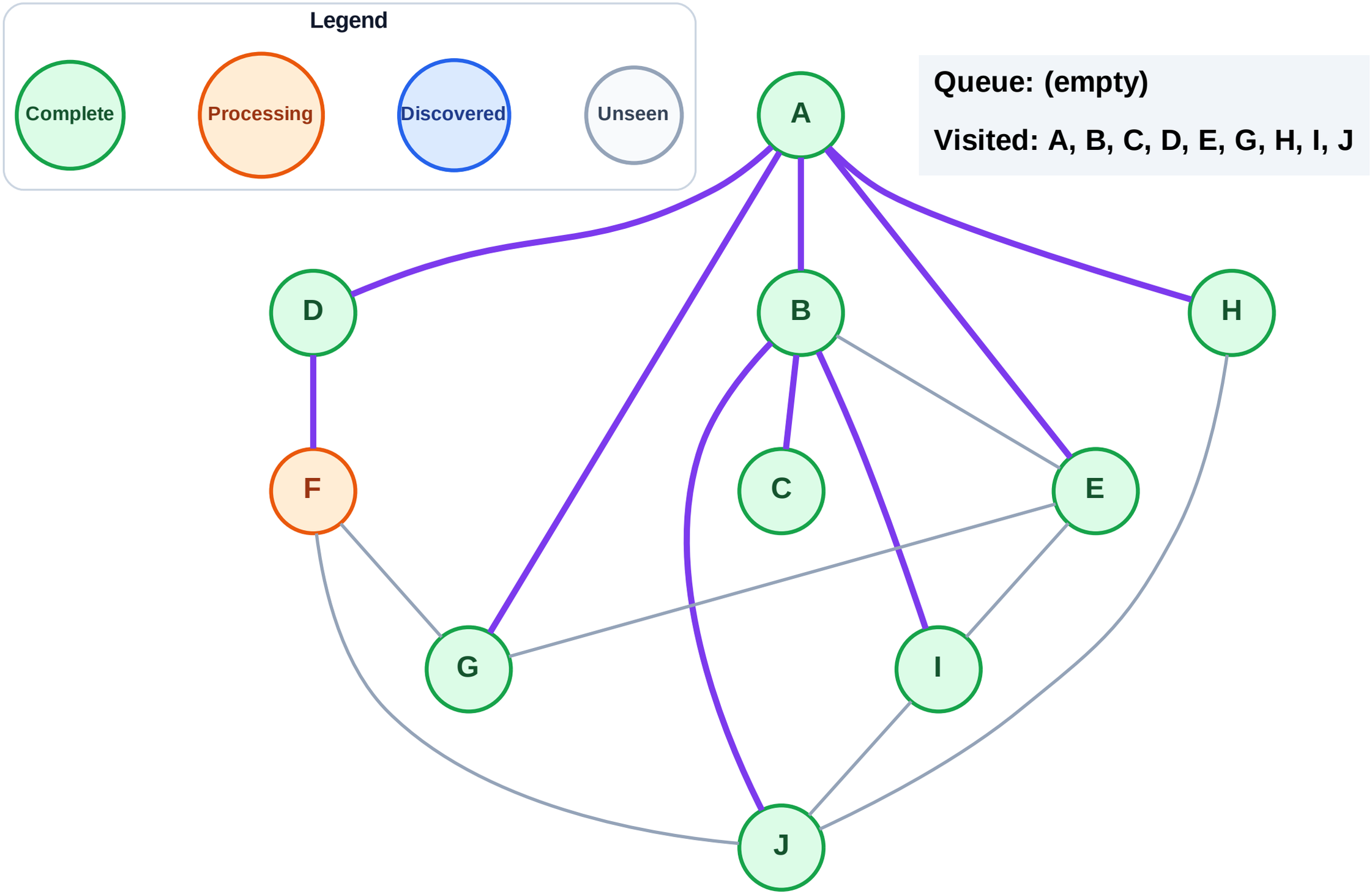
BFS Traversal

Step 19: No new neighbors from J



BFS Traversal

Step 20: Process node F

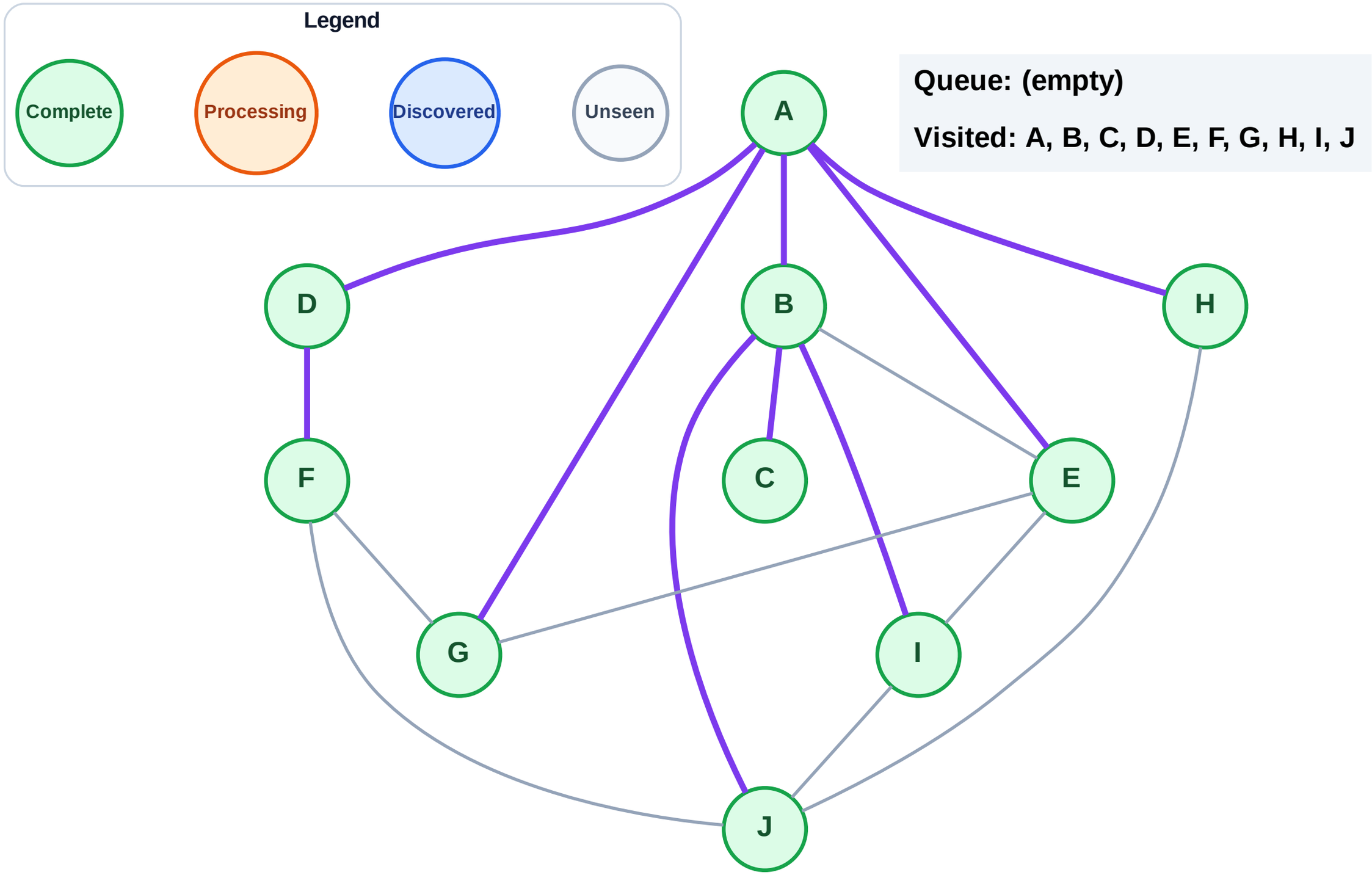


Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J

BFS Traversal

Step 21: No new neighbors from F

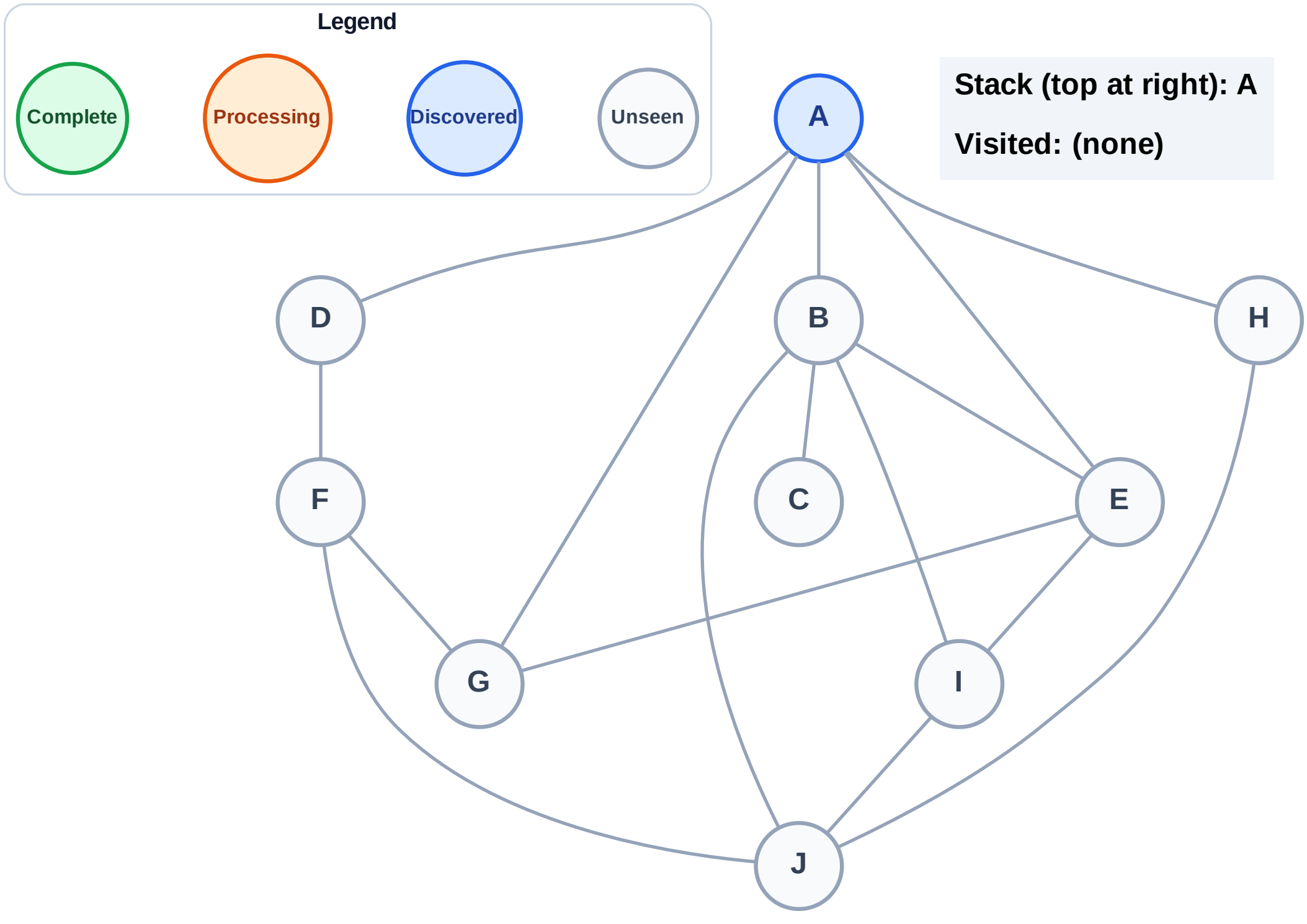


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

Step 1: Start at A

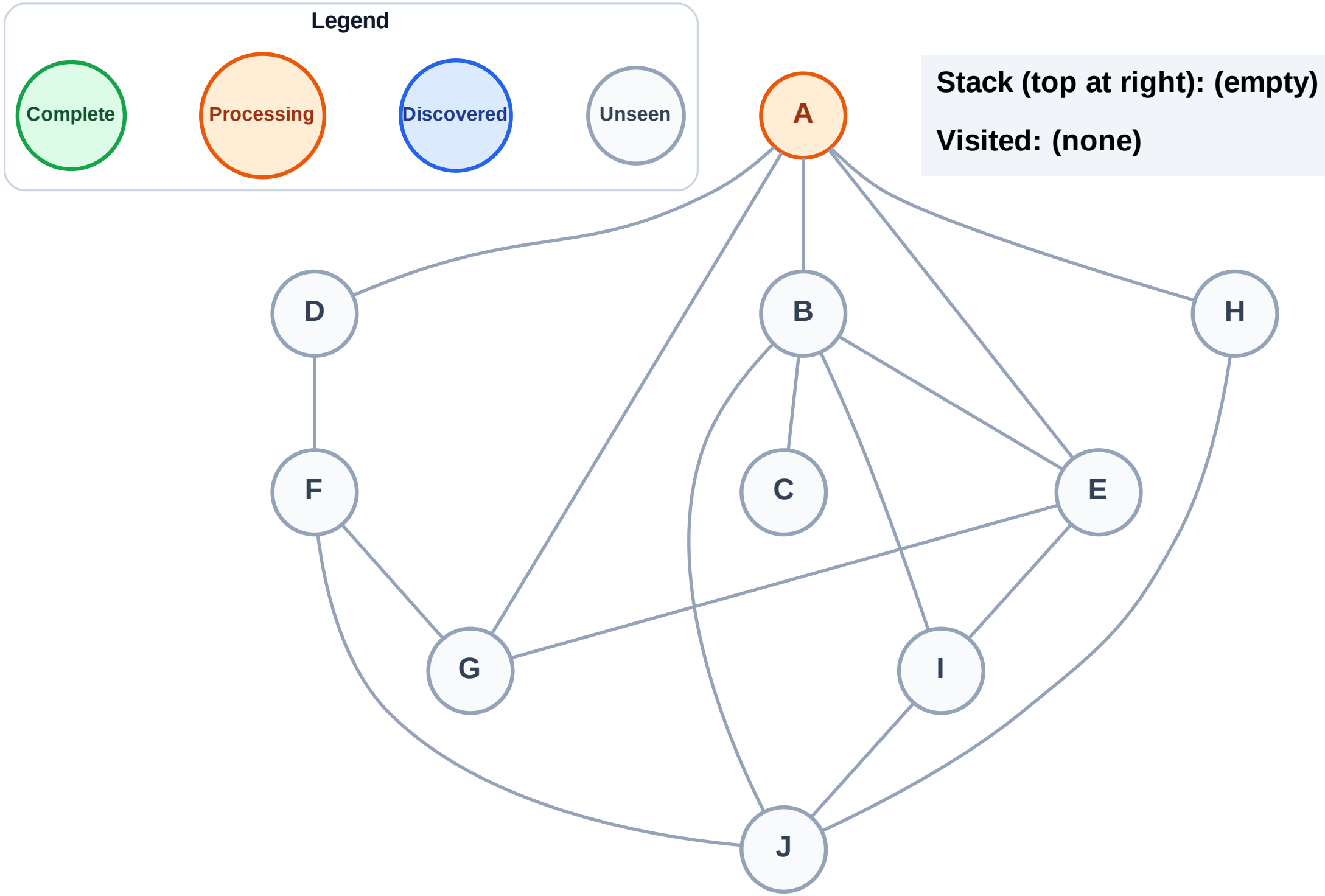


Stack (top at right): A

Visited: (none)

DFS Traversal

Step 2: Process node A

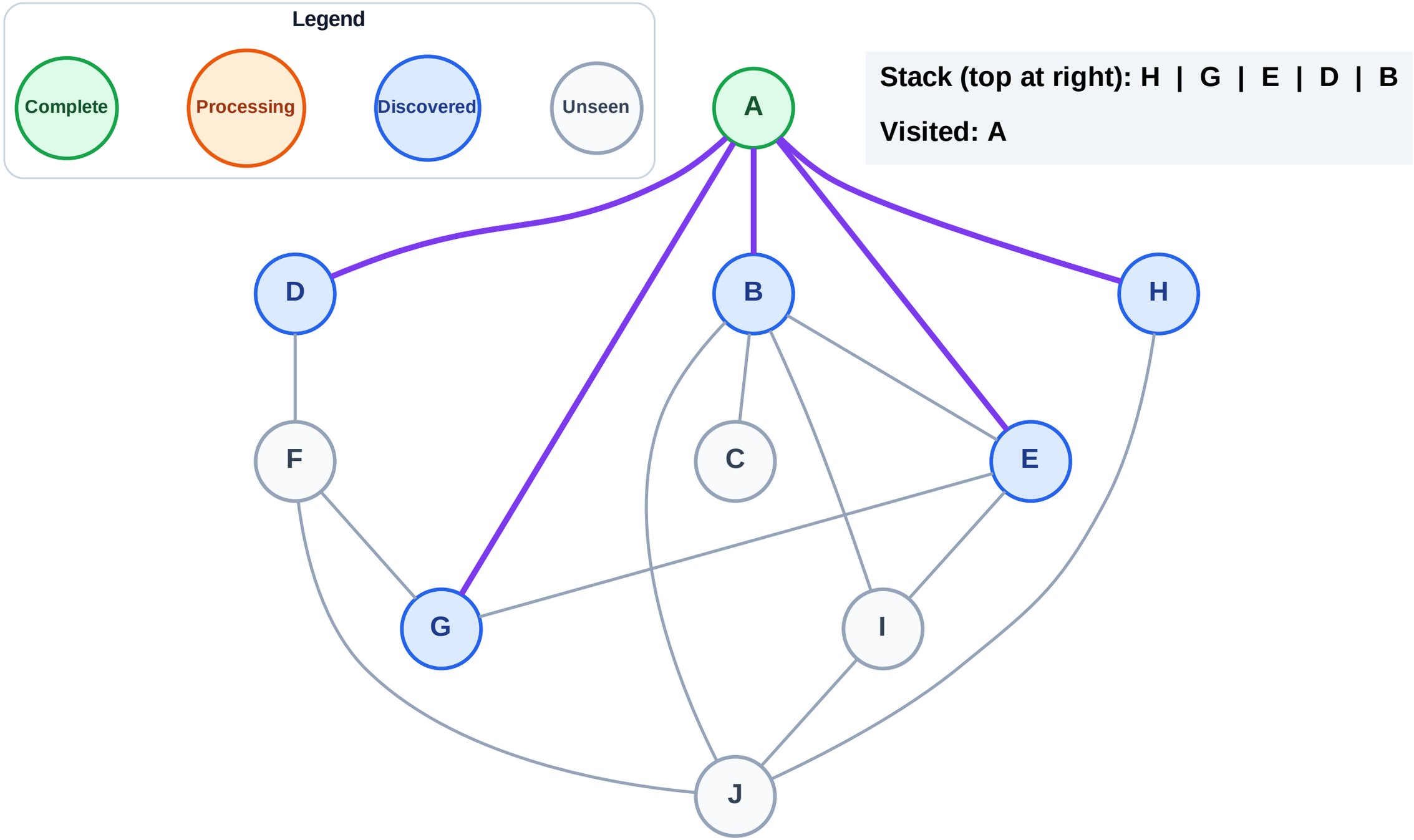


Stack (top at right): (empty)

Visited: (none)

DFS Traversal

Step 3: Discover H, G, E, D, B from A

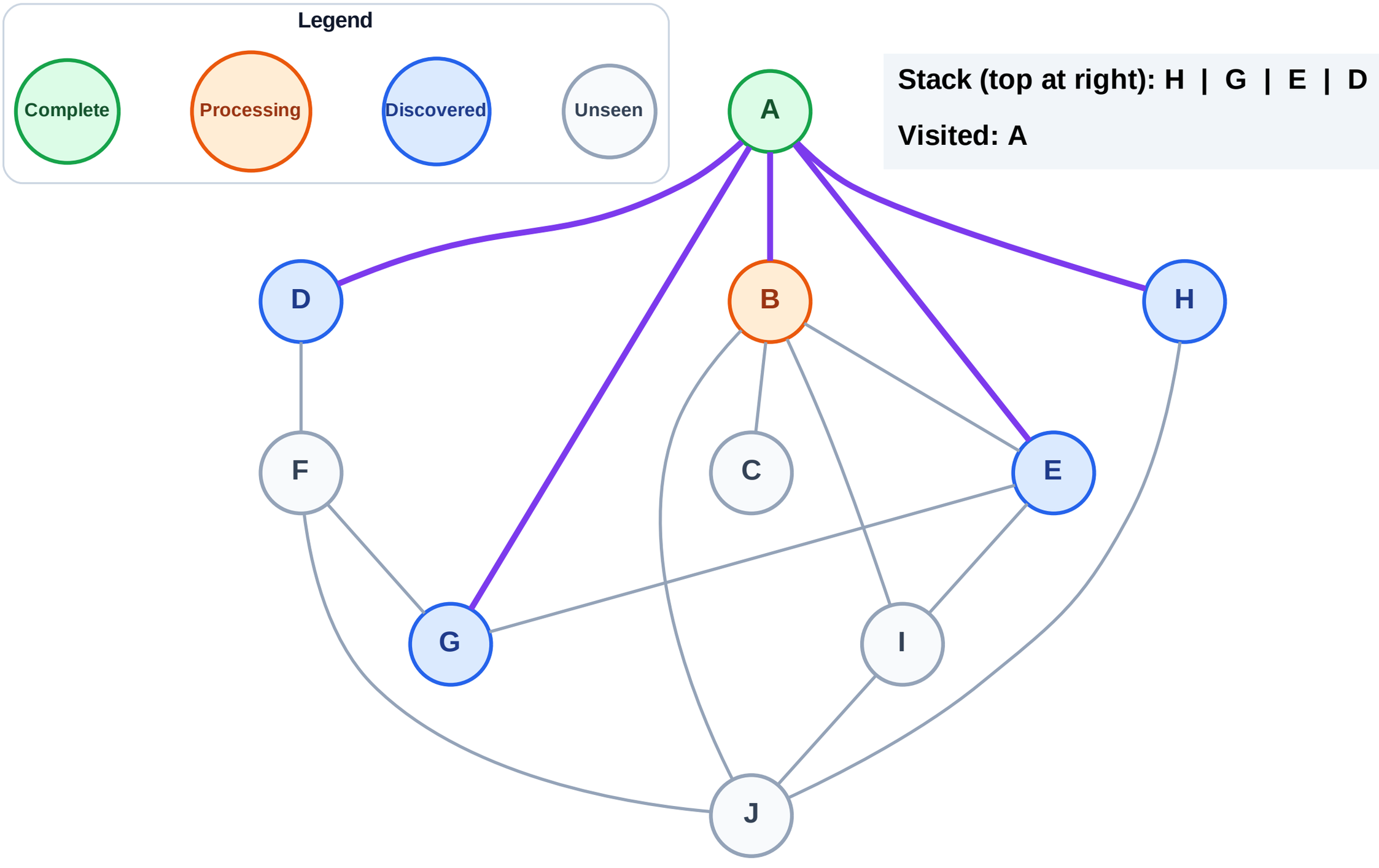


Stack (top at right): H | G | E | D | B

Visited: A

DFS Traversal

Step 4: Process node B

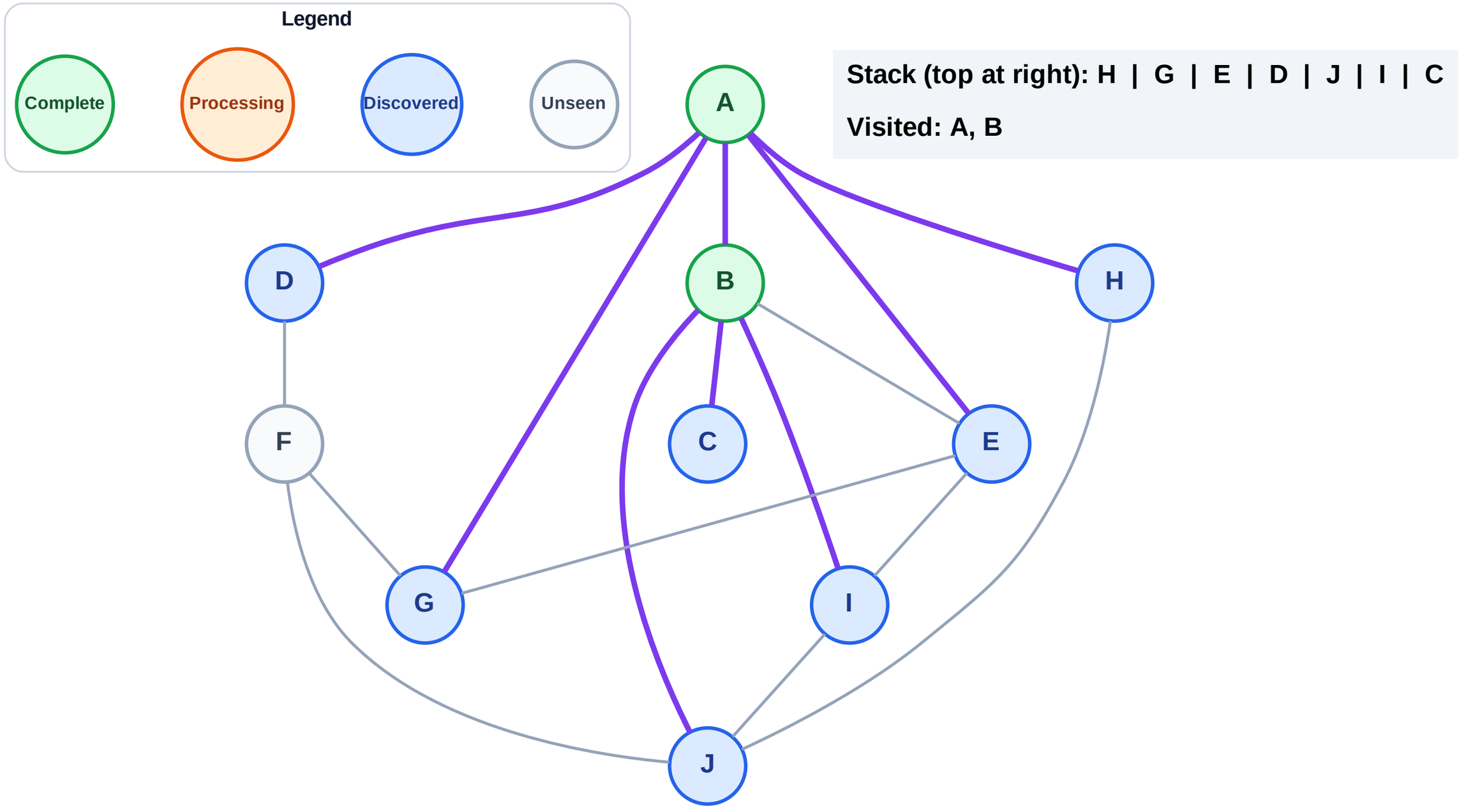


Stack (top at right): H | G | E | D

Visited: A

DFS Traversal

Step 5: Discover J, I, C from B

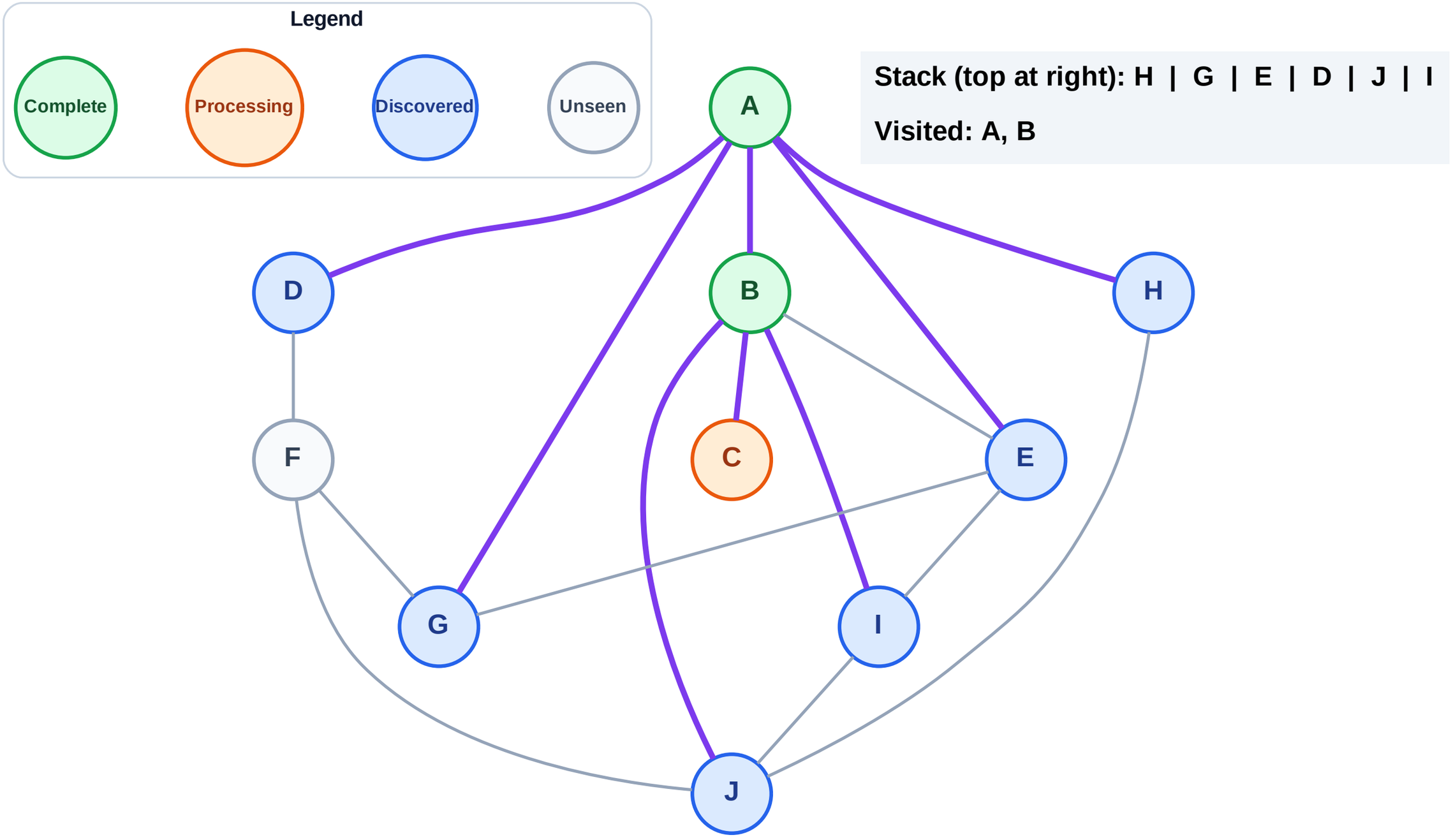


Stack (top at right): H | G | E | D | J | I | C

Visited: A, B

DFS Traversal

Step 6: Process node C

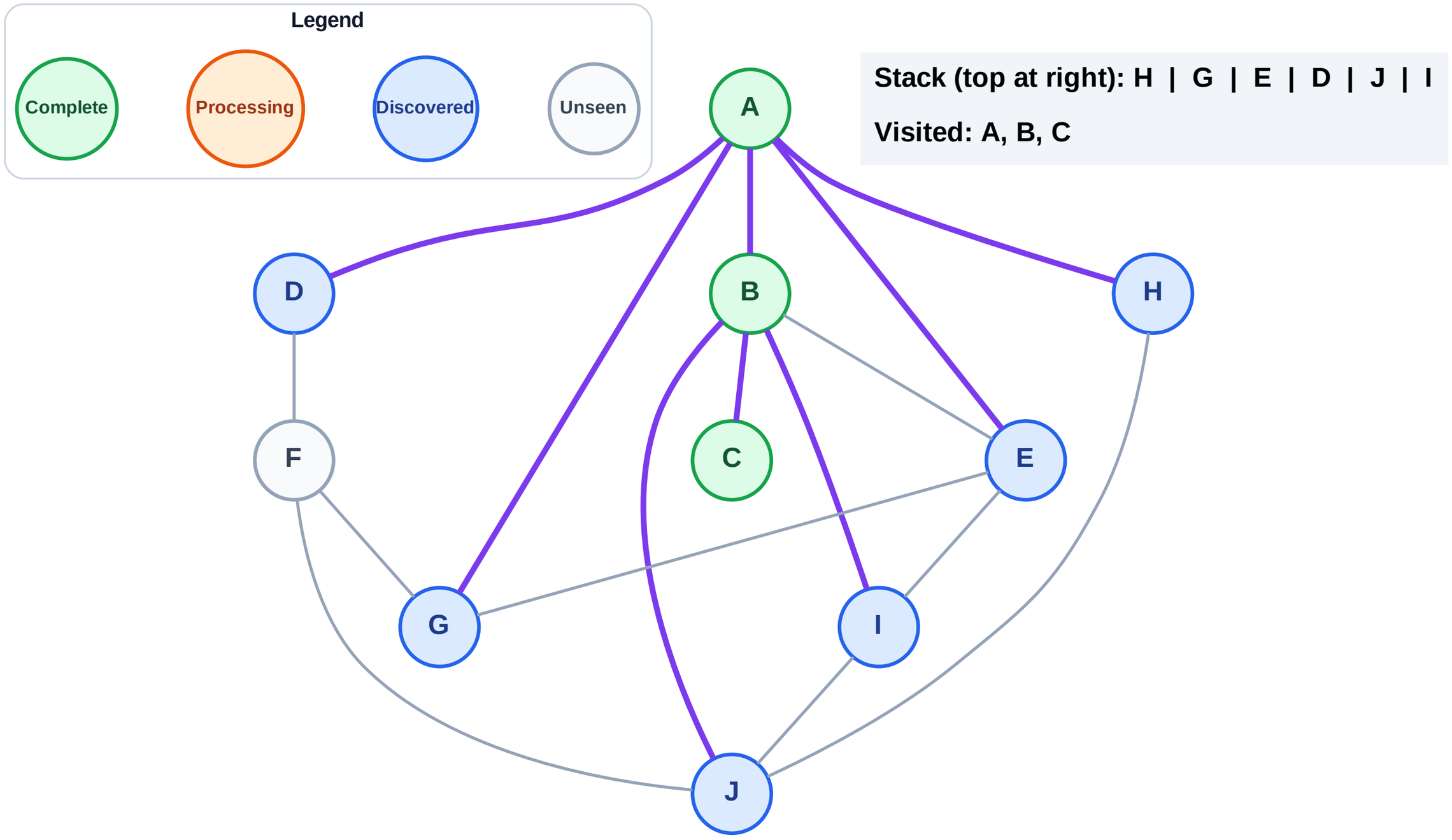


Stack (top at right): H | G | E | D | J | I

Visited: A, B

DFS Traversal

Step 7: No new neighbors from C

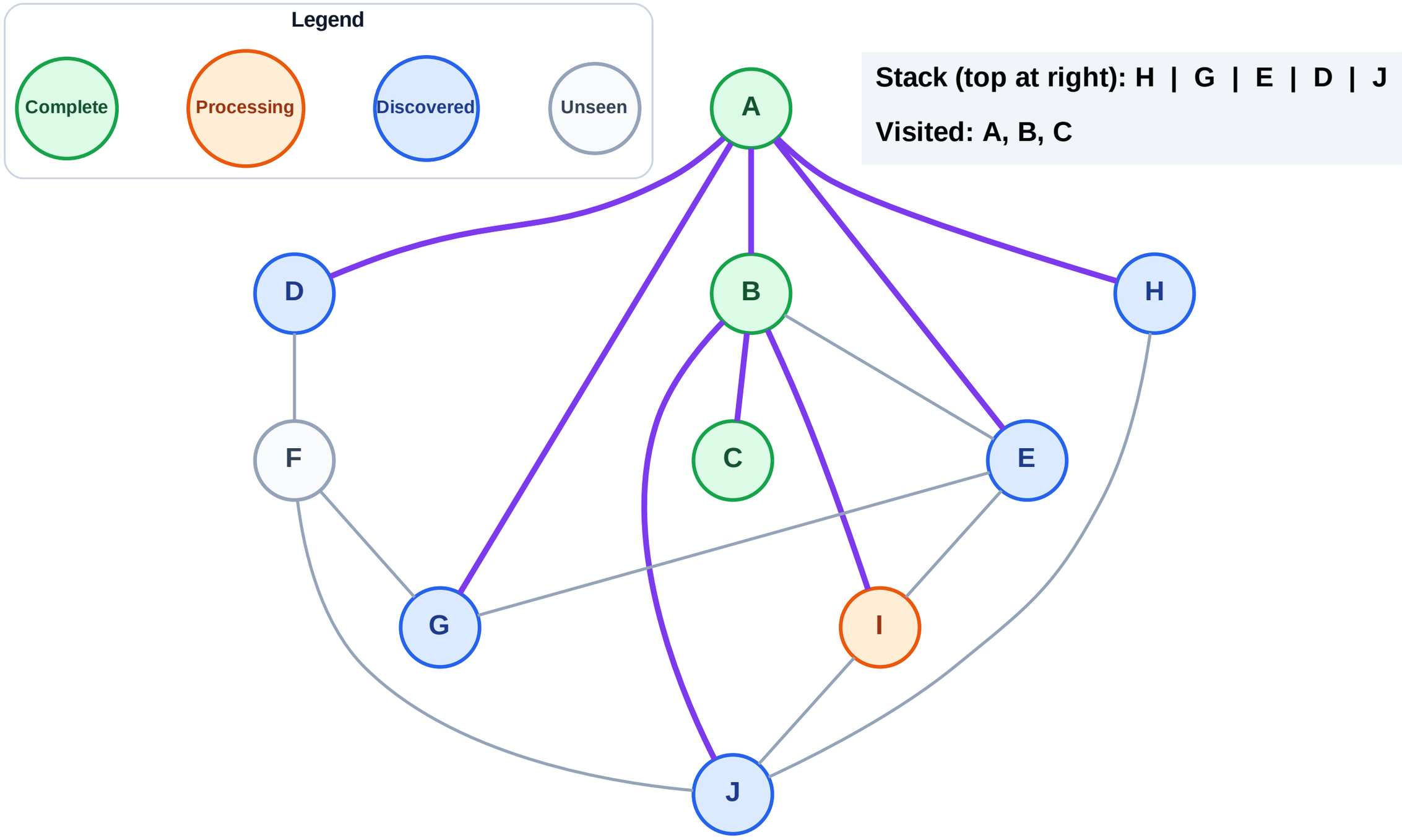


Stack (top at right): H | G | E | D | J | I

Visited: A, B, C

DFS Traversal

Step 8: Process node I

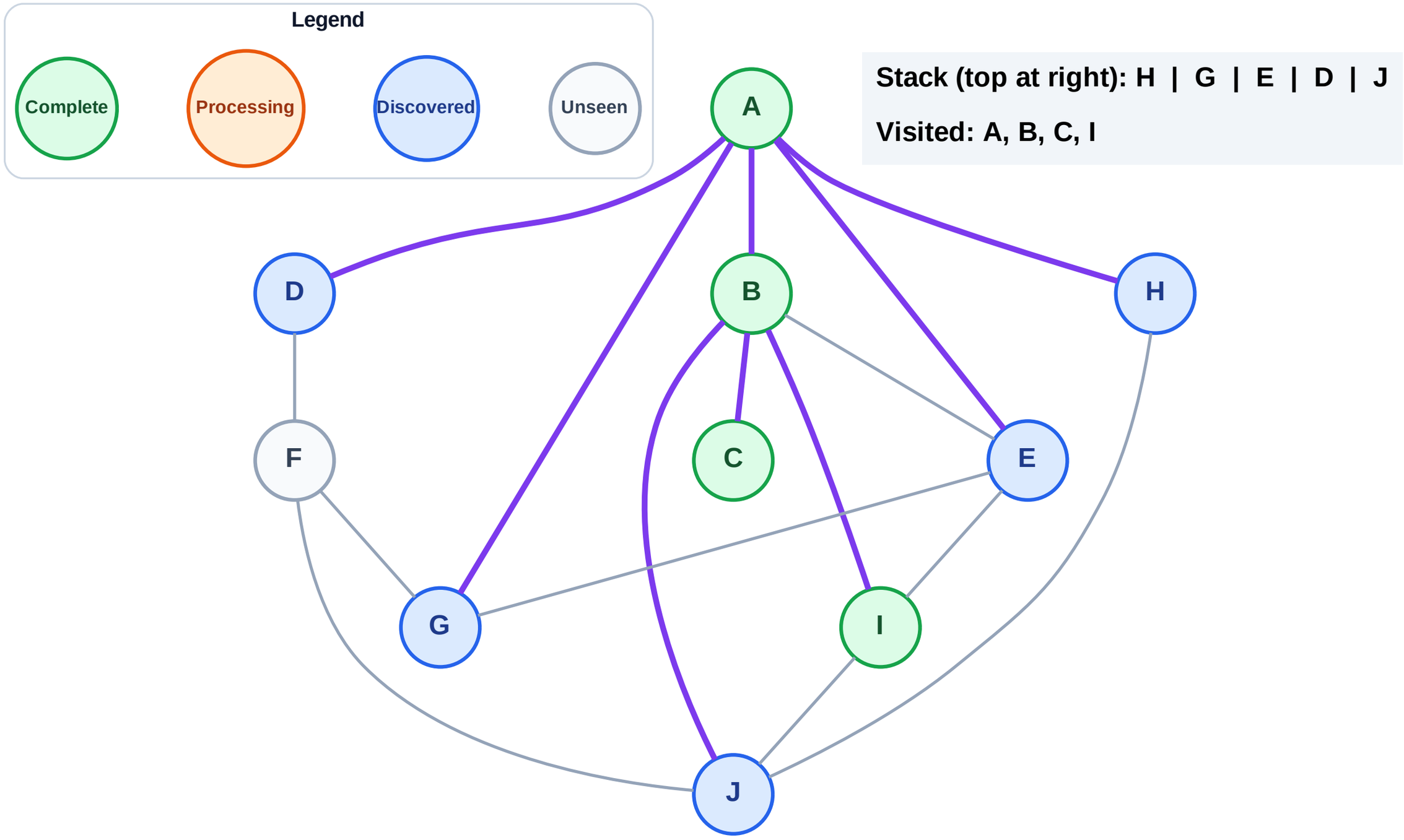


Stack (top at right): H | G | E | D | J

Visited: A, B, C

DFS Traversal

Step 9: No new neighbors from I

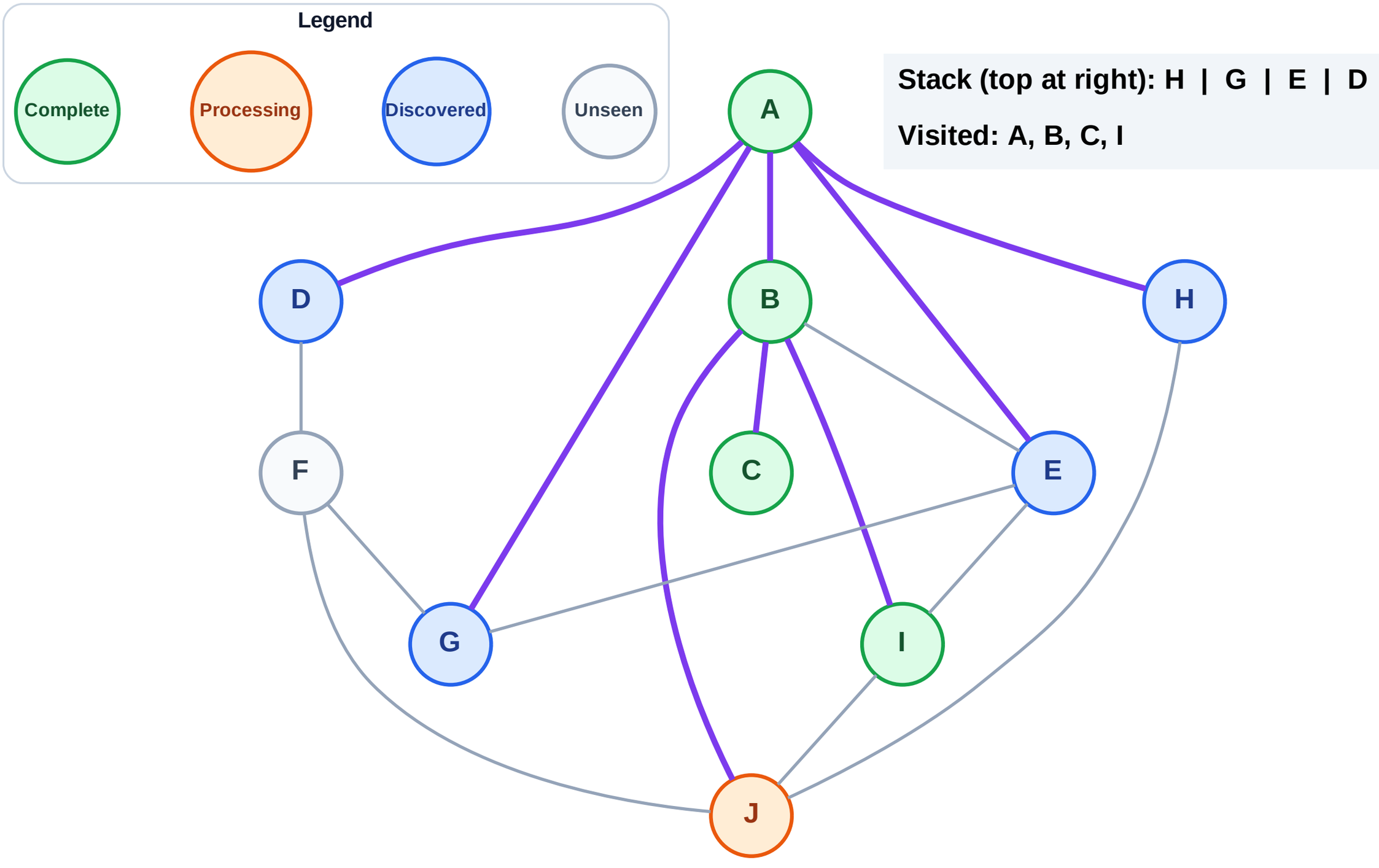


Stack (top at right): H | G | E | D | J

Visited: A, B, C, I

DFS Traversal

Step 10: Process node J

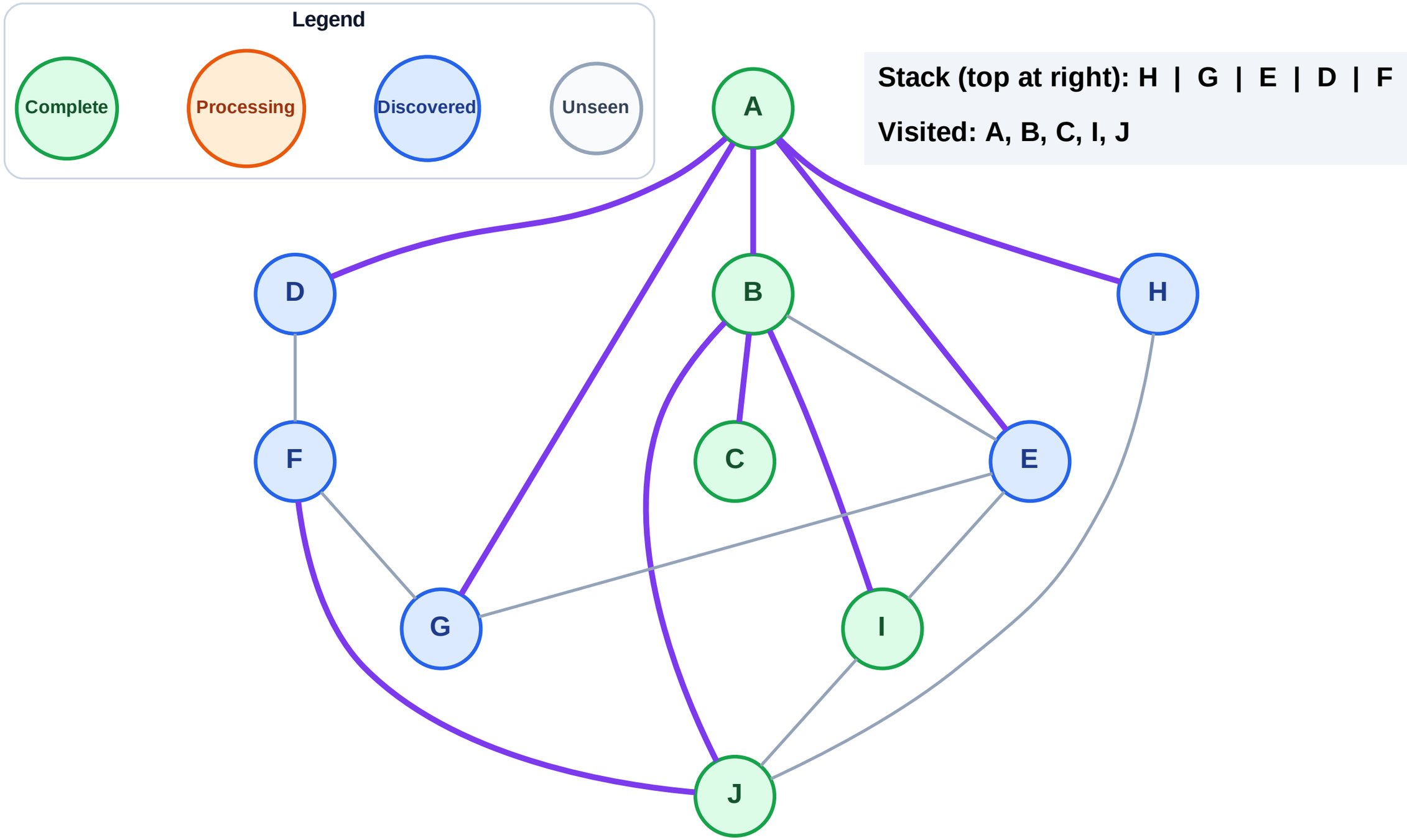


Stack (top at right): H | G | E | D

Visited: A, B, C, I

DFS Traversal

Step 11: Discover F from J

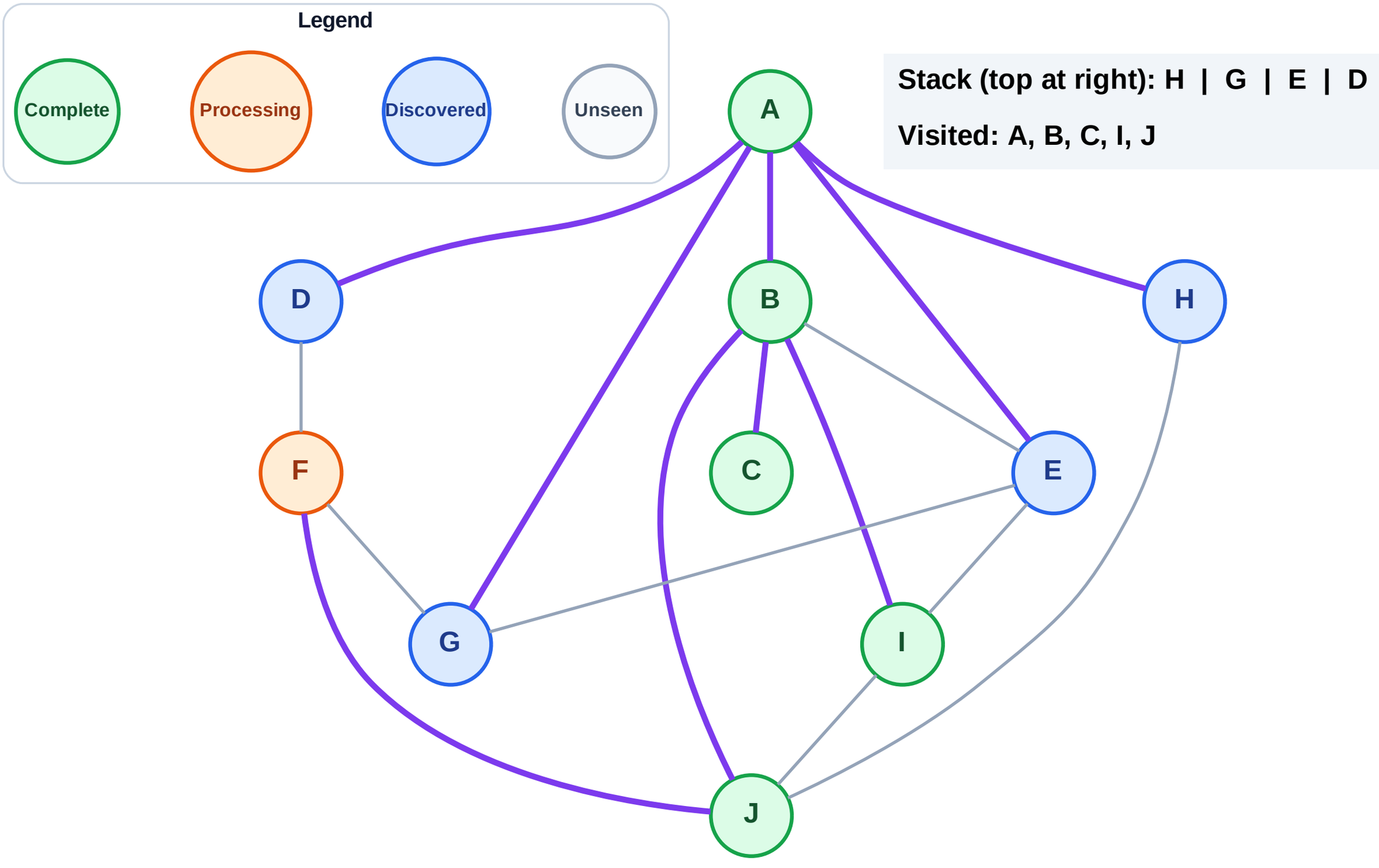


Stack (top at right): H | G | E | D | F

Visited: A, B, C, I, J

DFS Traversal

Step 12: Process node F

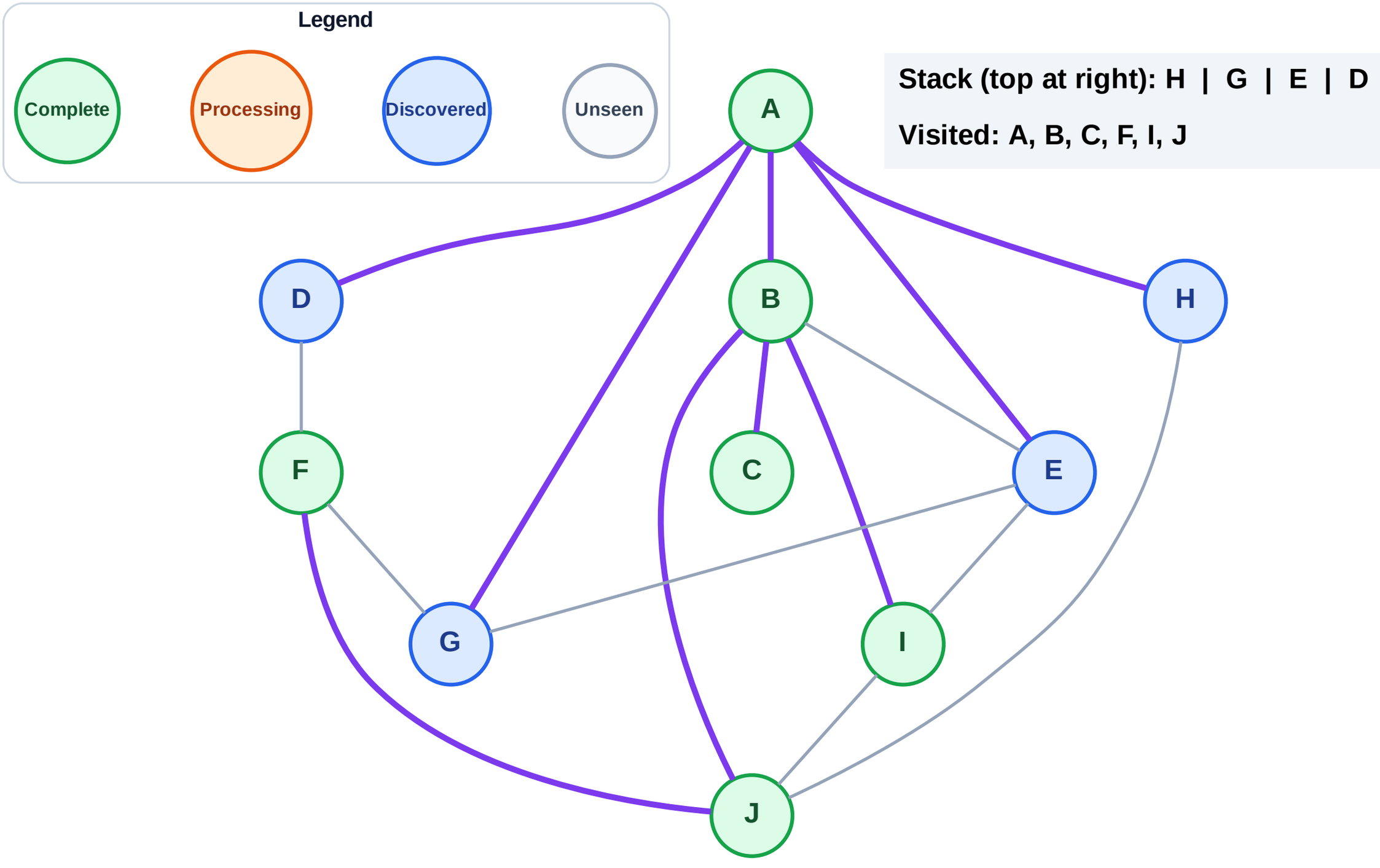


Stack (top at right): H | G | E | D

Visited: A, B, C, I, J

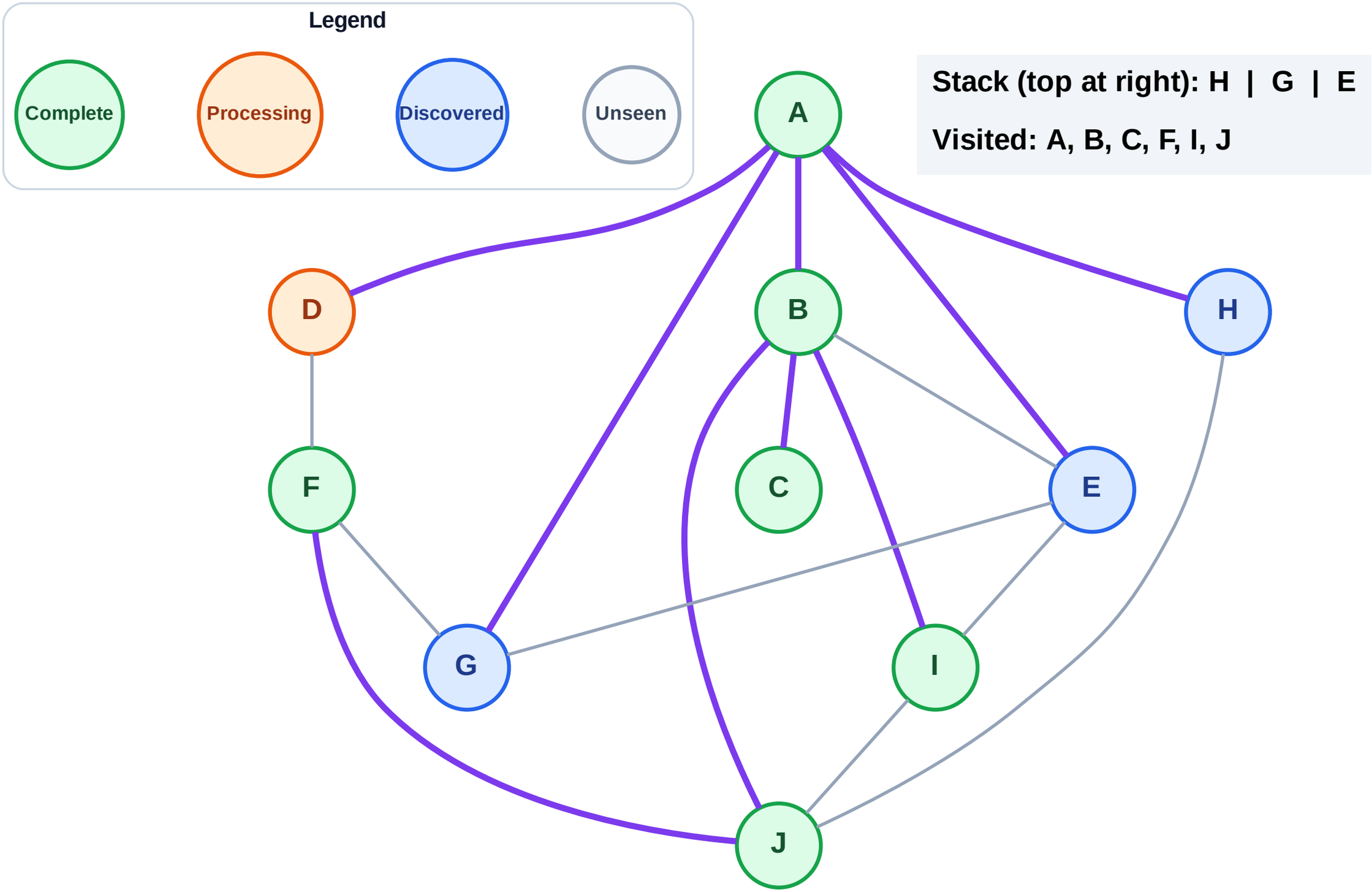
DFS Traversal

Step 13: No new neighbors from F



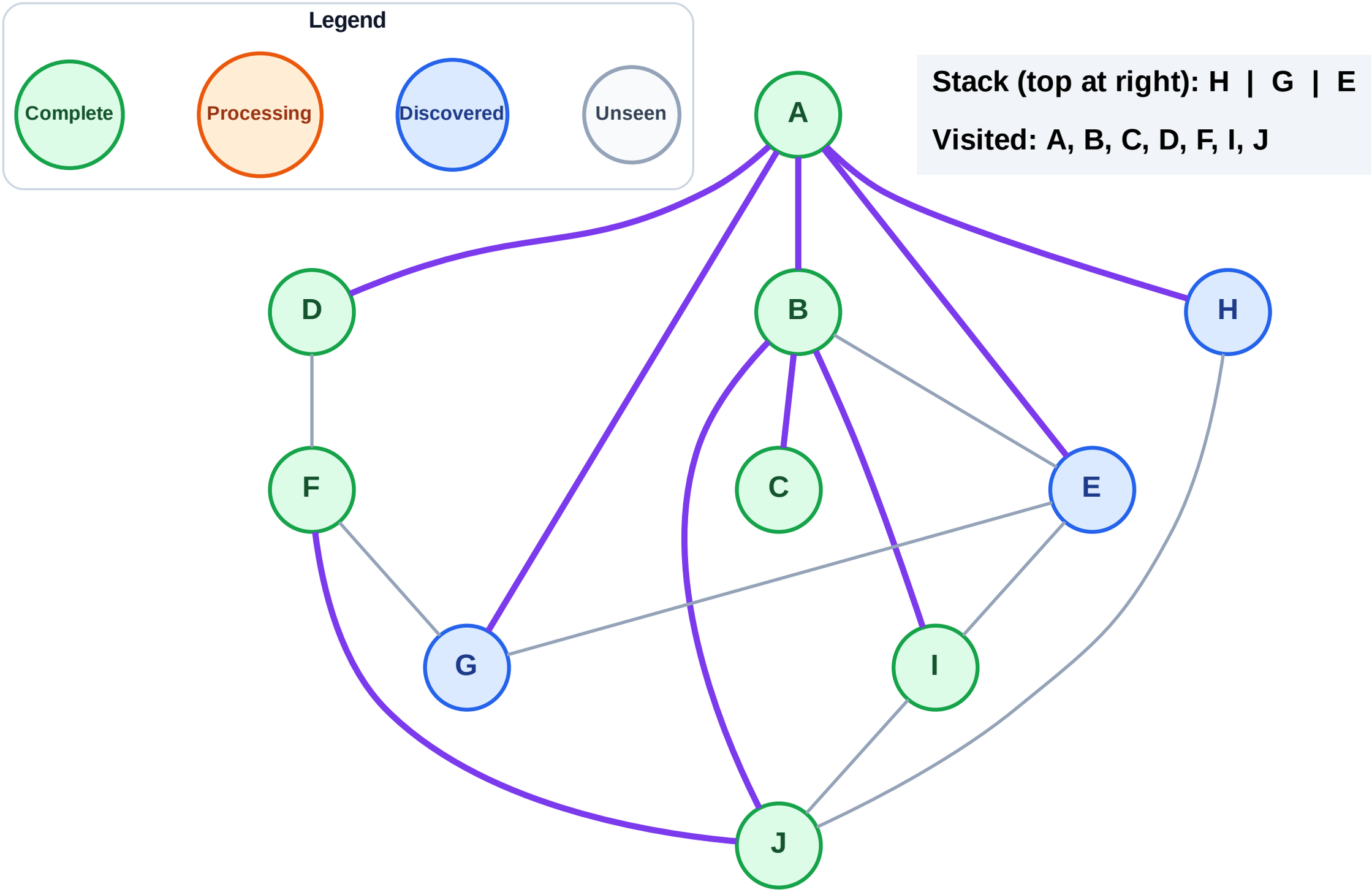
DFS Traversal

Step 14: Process node D



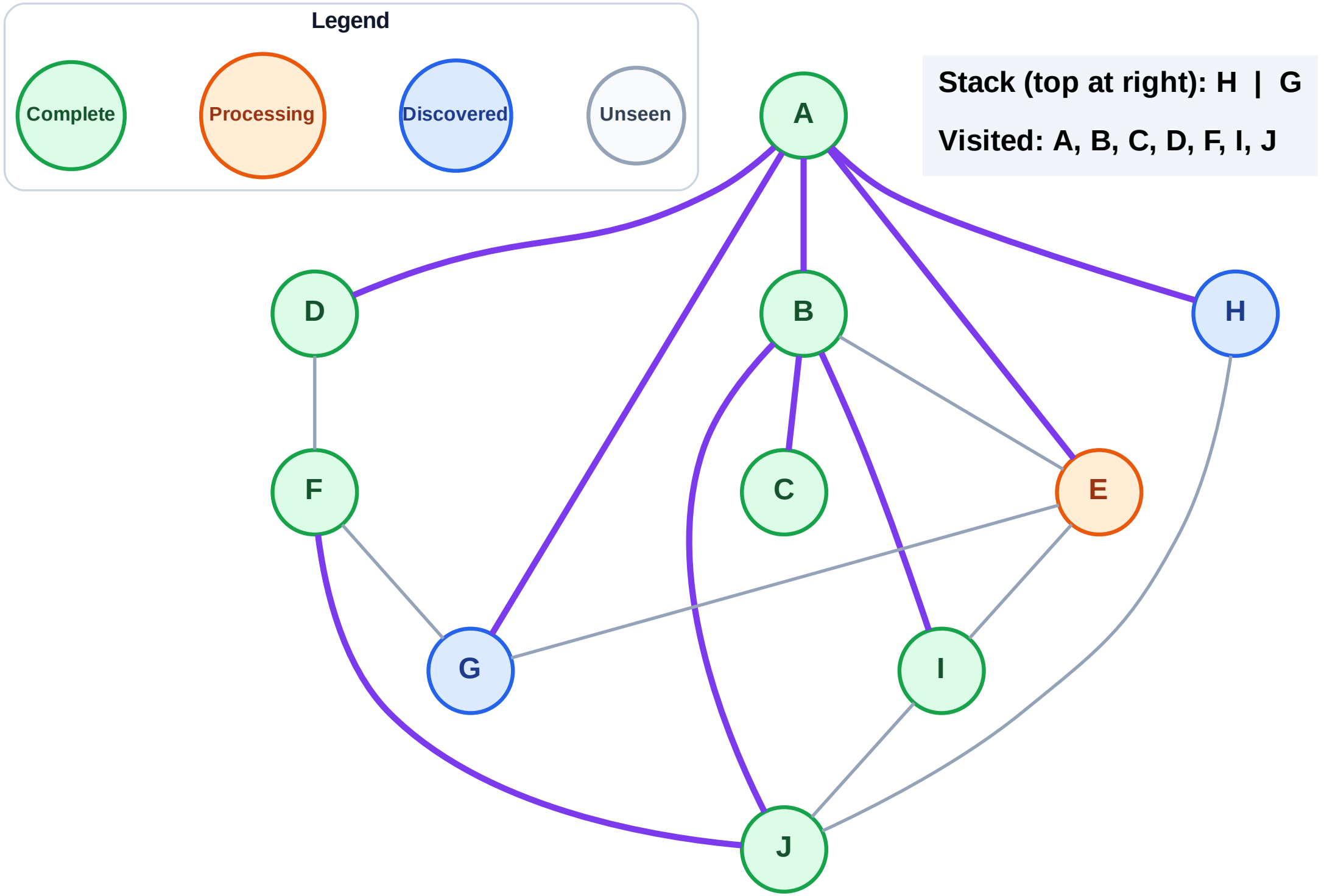
DFS Traversal

Step 15: No new neighbors from D



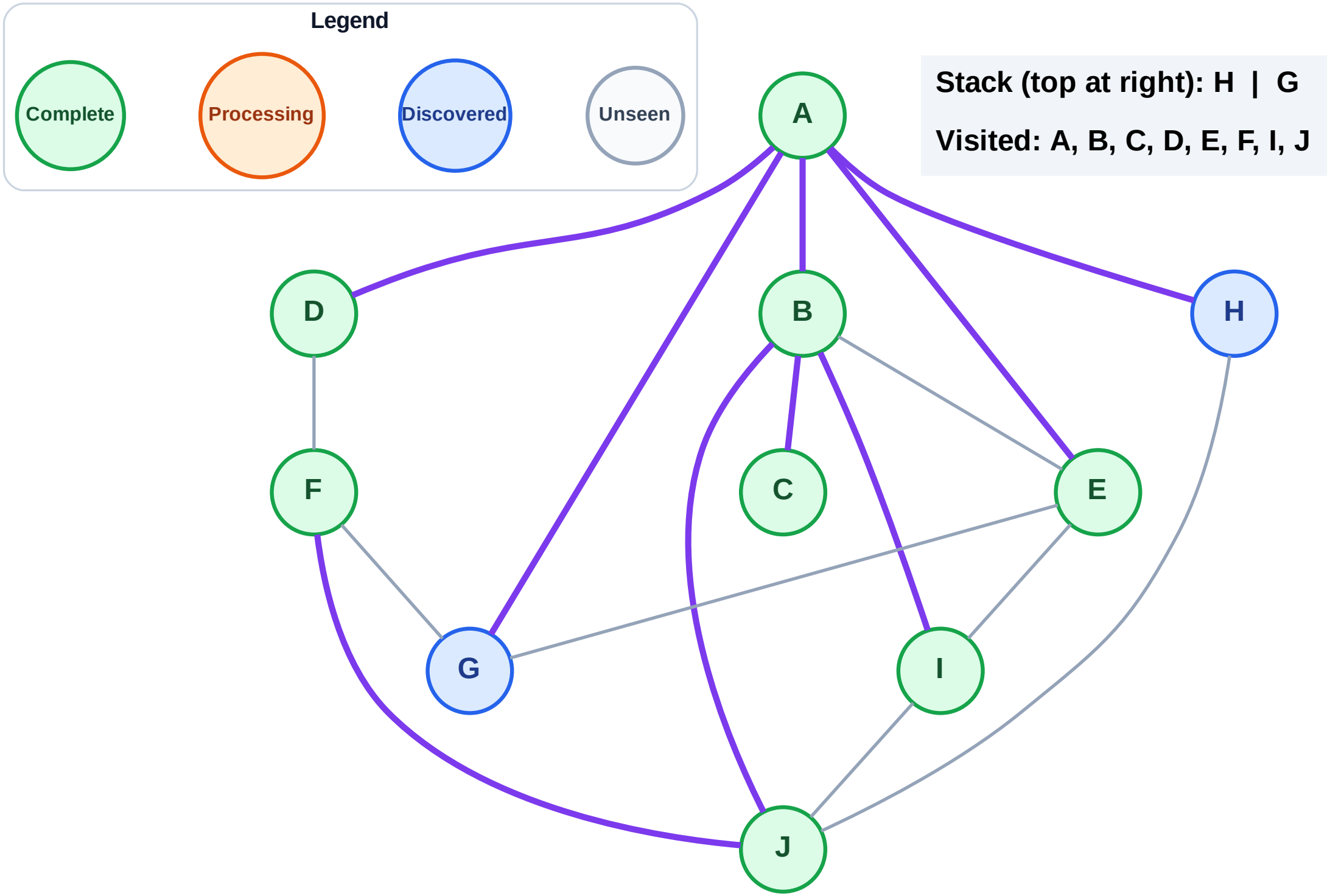
DFS Traversal

Step 16: Process node E



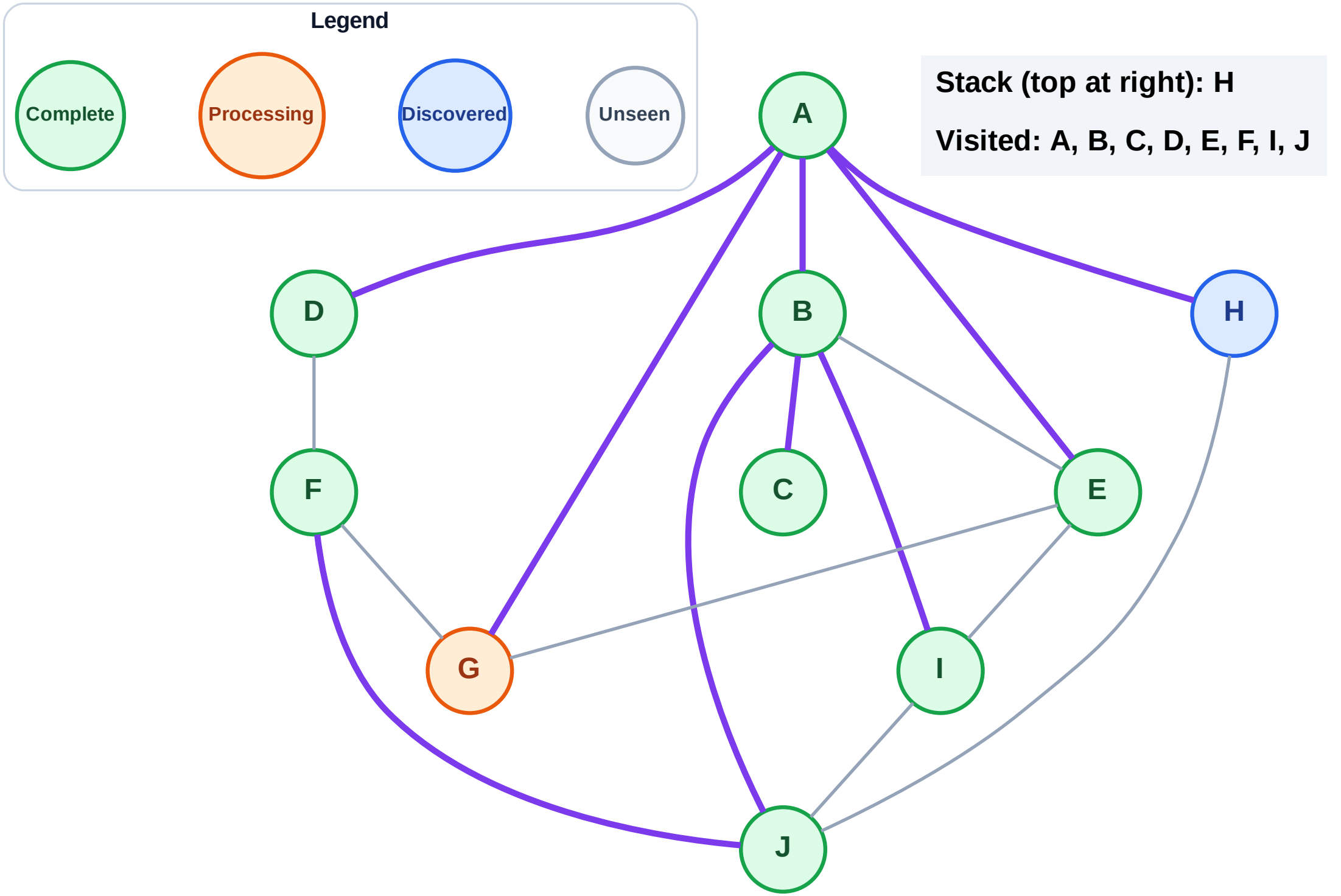
DFS Traversal

Step 17: No new neighbors from E



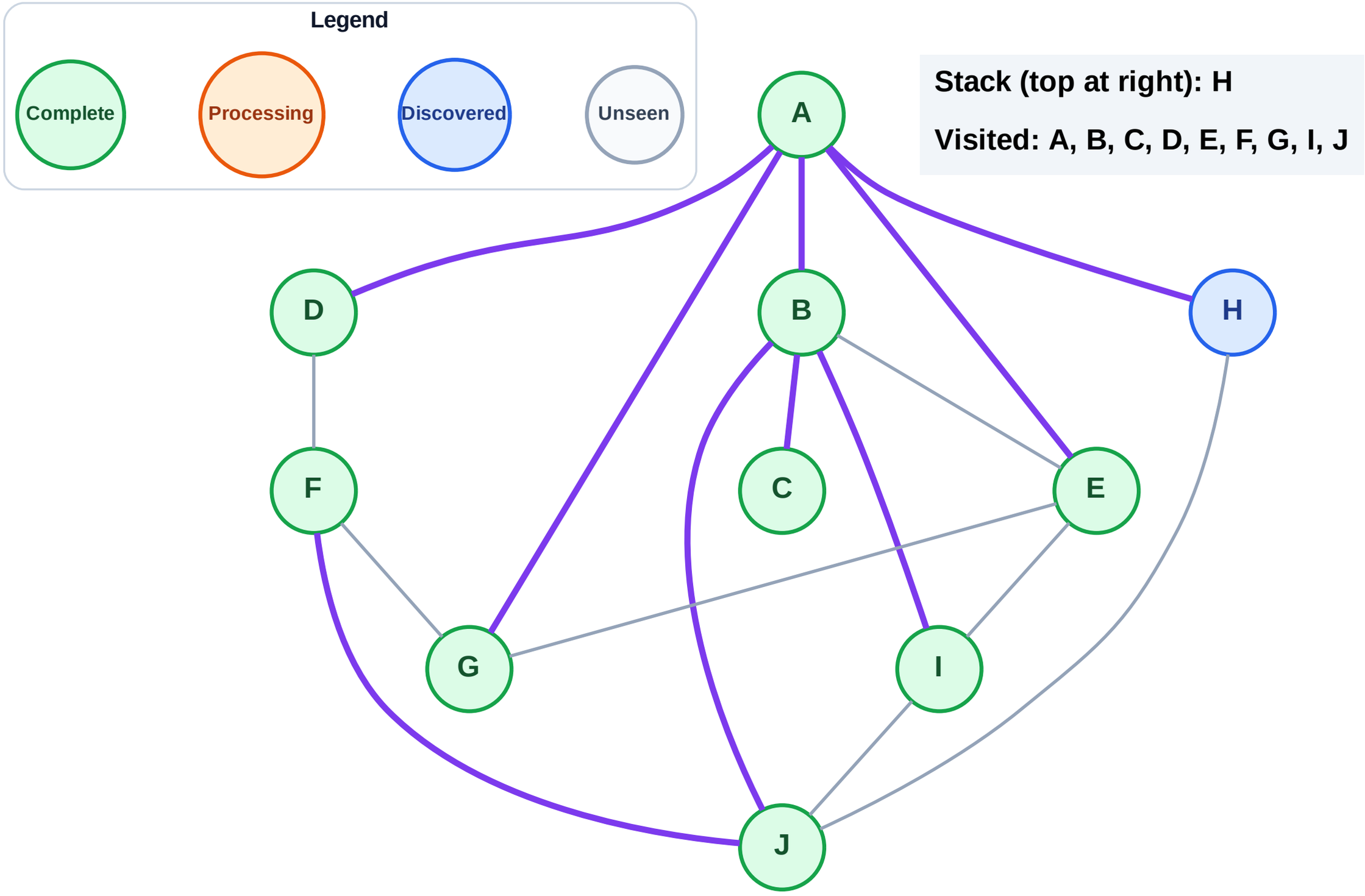
DFS Traversal

Step 18: Process node G



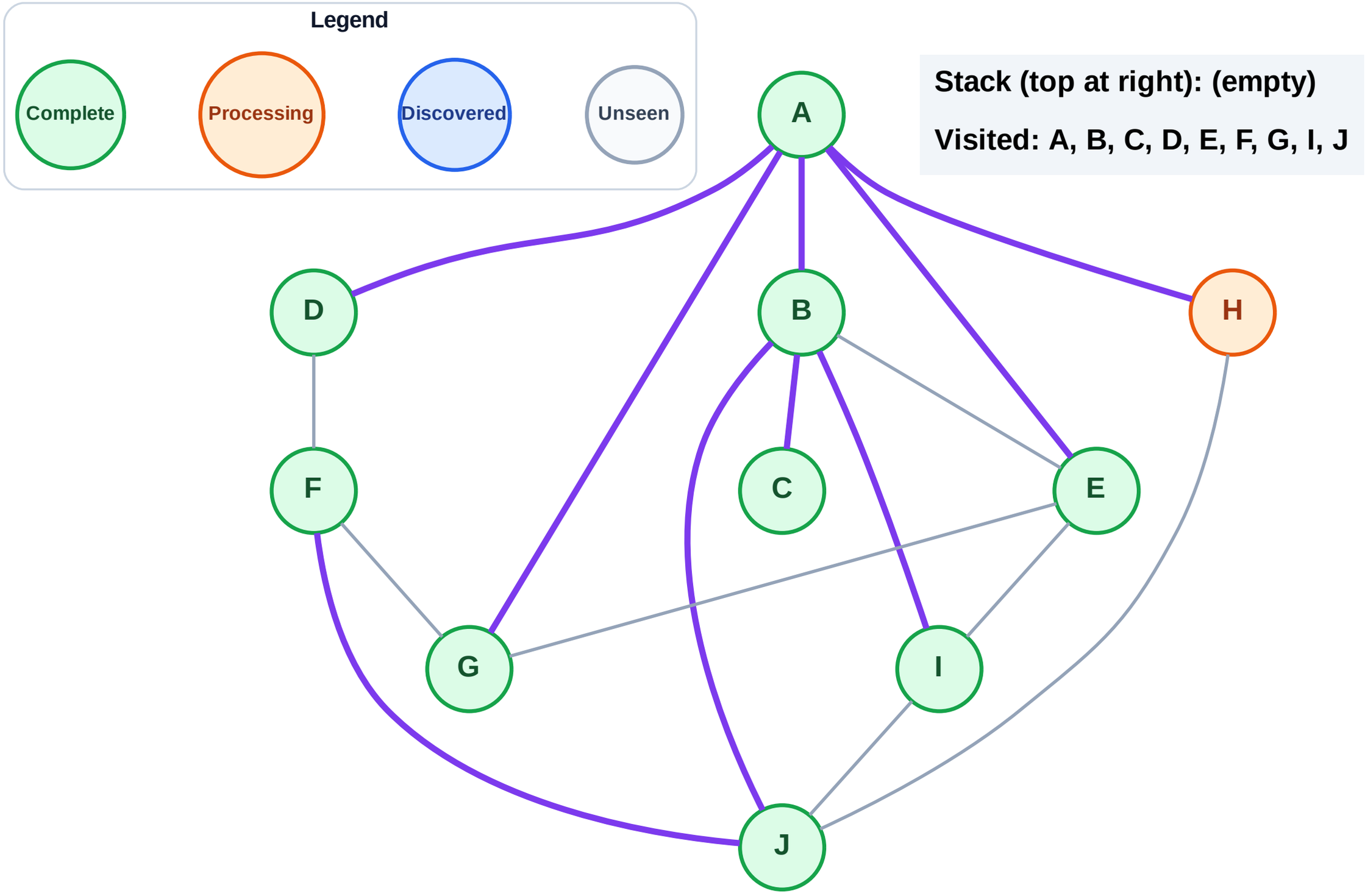
DFS Traversal

Step 19: No new neighbors from G



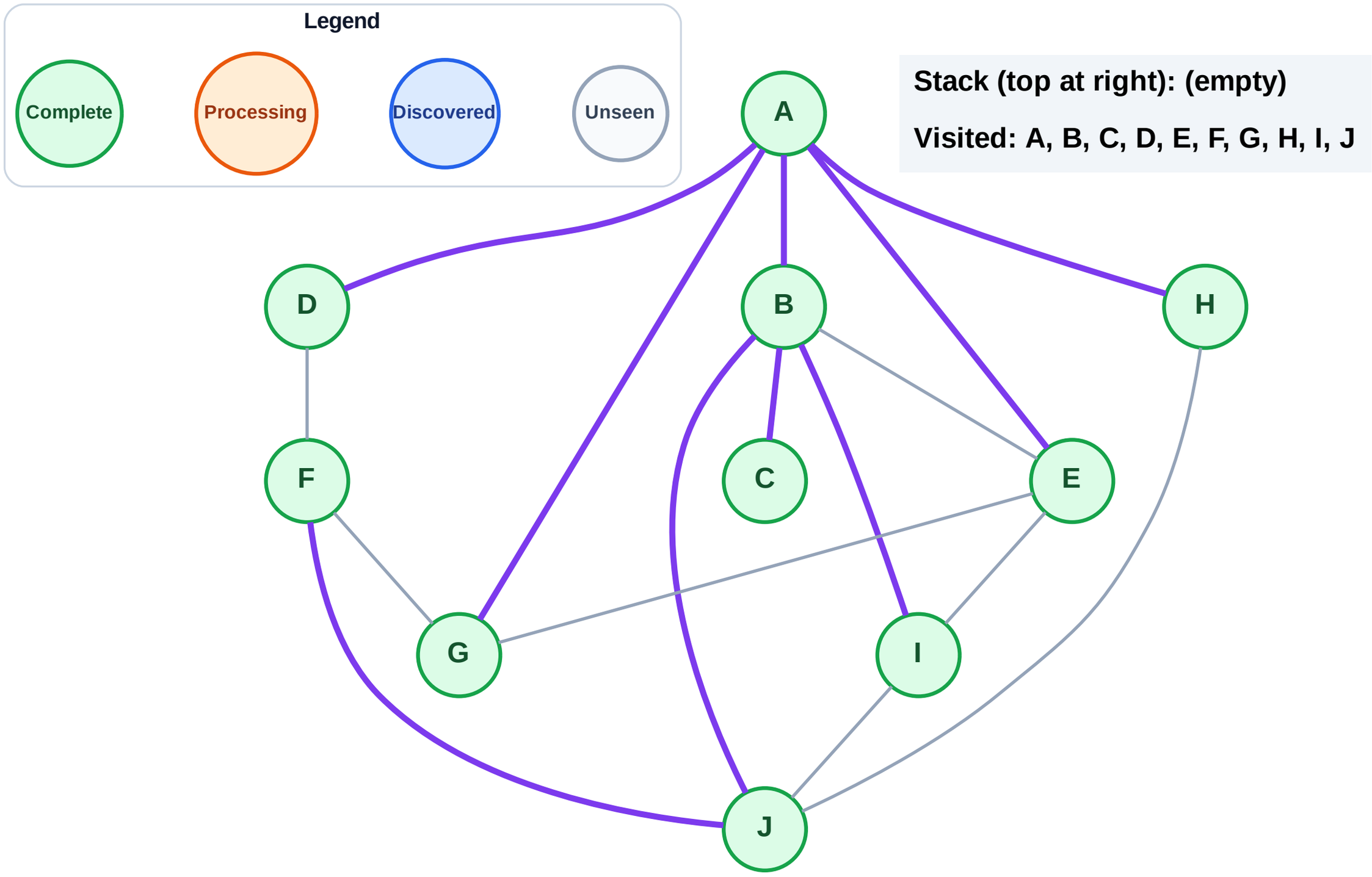
DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J